DI32015 - Manufacturing Project II

Lab Report

Group 004

Ultraviolet Photodetection Devices

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Part I	Part II	Part III	Part IV	Part V	Part VI



Abstract

This report introduces and concludes the manufacture and test of an ultraviolet photodetection device. Under the background of the application of ultraviolet light, this experiment a Ni/n-ZnO/Ni horizontal structure ultraviolet photodetector was prepared, a single-layer ZnO film was sputtered on a sapphire substrate by magnetron sputtering, the interdigital electrode was etched by ultraviolet lithography technology, and the Ni electrode was sputtered under vacuum by magnetron sputtering. Subsequently, the photoelectric performance of the traditional ZnO ultraviolet detector was tested and analysed, and the volt-ampere characteristic curves, the responsivity at different wavelengths, the switching time of the device under five illumination cycles, and the comparison of the optical response time under different bias voltages were studied. The current-voltage (I-V) curves of a horizontally structured Ni/n-ZnO/Ni photodetector should exhibit symmetrical and near-linear I-V characteristics under low light and 360 nm light irradiation, while exhibiting the same photo response characteristics at positive and negative biases. At the same time, under the irradiation of low light of different wavelengths from 320nm to 400nm, the scanning voltage range is -3~3V, and a symmetrical and almost linear I-V curve is observed, which is consistent with the ultraviolet light response characteristics. By analysing the responsivity at different wavelengths, the cut-off wavelength of the device can be calculated, which is 440 nm. The device is illuminated for 5 seconds and removed for five cycles, with the current increasing as the light is irradiated and then decreasing as the light is turned off. It is concluded that the shorter the response time, the more sensitive the UV photodetector is. Under four different bias voltages of 0.5V, 1V, 1.5V and 2V, it is concluded that the large voltage is conducive to the migration of photogenerated carriers and causes a large change in photocurrent.

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I. Introduction

1) Experimental Background

- **Phylogeny of ultraviolet light**

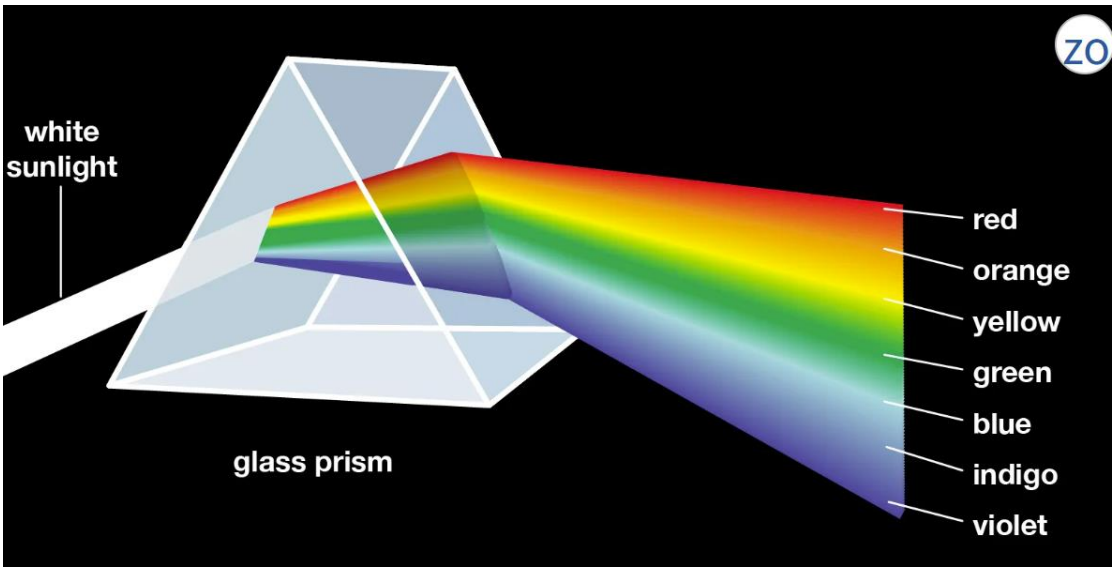


Figure 1 Isaac Newton's prism experiment [1]

In 1666, Newton expanded the sunlight through a glass prism into a spectrum of various colours from red to violet, and he discovered that white light was composed of various colours of light, but this was only the discovery of the visible spectrum (Fig. 1).

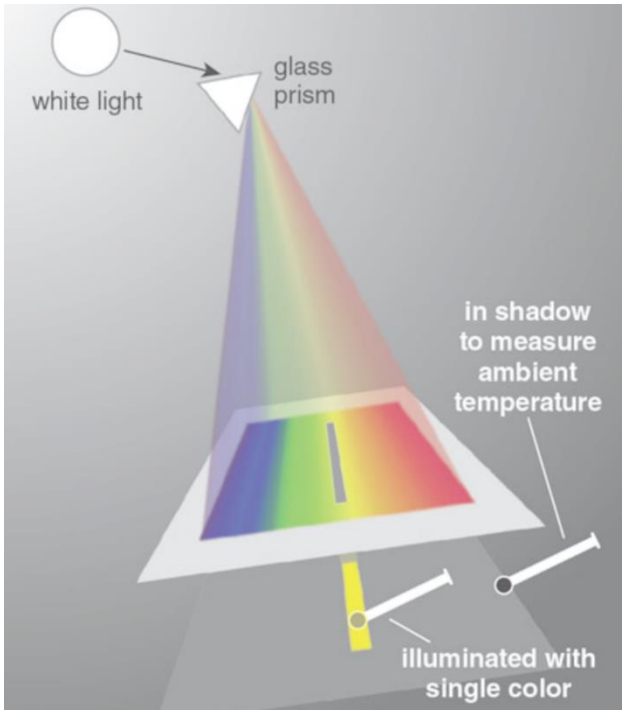


Figure 2 Herschel and far-infrared heating [2]

In 1800, the famous British astronomer William Herschel was studying the heating effect of the solar spectrum and accidentally discovered that the shadow area outside the red light could greatly

increase the number of thermometers (Fig. 2). He believed that outside of red light, there must be some kind of invisible ray, that is, infrared light.

In 1801, the German physicist and chemist Johann Wilhelm Ritter discovered that in addition to the violet end of the visible spectrum, there is also invisible light. Ritter soaked a piece of paper in a silver chloride solution near the violet region of the prism's visible spectrum, and he noticed that the outer area of the violet light was strongly blackened, indicating that this part was irradiated by invisible radiation. Ritter called this invisible light "oxygen rays", which became known as ultraviolet light.

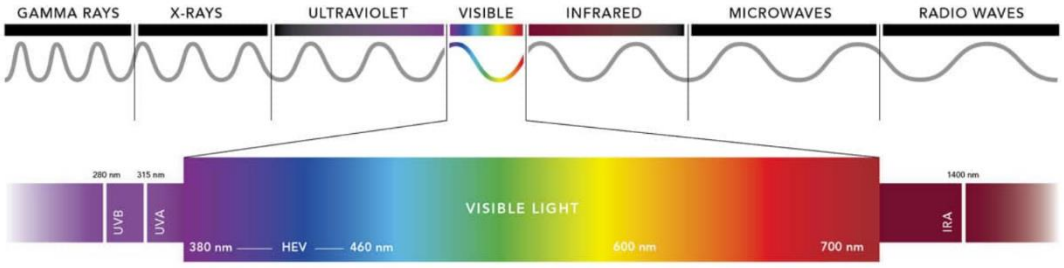


Figure 3 Sunlight spectrum [3]

Then other wavelengths of electromagnetic waves were discovered one after another. Until today, here is a summary: sunlight is a kind of polychromatic light, after passing through the dispersion system, it will form a spectrum as shown in the figure, from left to right, γ-rays, X-rays, ultraviolet rays, visible light(purple-blue-indigo-green-yellow-orange-red), infrared rays, microwaves and radio waves (Fig. 3).

Ultraviolet light, with a frequency range of 750 THz to 30 PHz in the electromagnetic spectrum and a vacuum wavelength of 400 nm to 10 nm, is a type of invisible light. As sunlight passes through the atmosphere, its ultraviolet component is significantly weakened, resulting in three distinct types of UV light that experience varying degrees of atmospheric attenuation. UVA, characterized by low frequency and long wavelengths (400 nm to 320 nm), is not absorbed by the ozone layer in the Earth's stratosphere, allowing approximately 90% to reach the surface due to its strong penetrating ability. UVB, with medium frequency and medium wavelengths (320 nm to 280 nm), is mostly absorbed by stratospheric ozone, with only about 10% reaching the ground, though it retains some penetration capability. UVC, the highest-frequency and shortest-wavelength type (280 nm to 100 nm), is almost entirely absorbed by stratospheric ozone, making it the least penetrating form of ultraviolet light (Fig. 4).

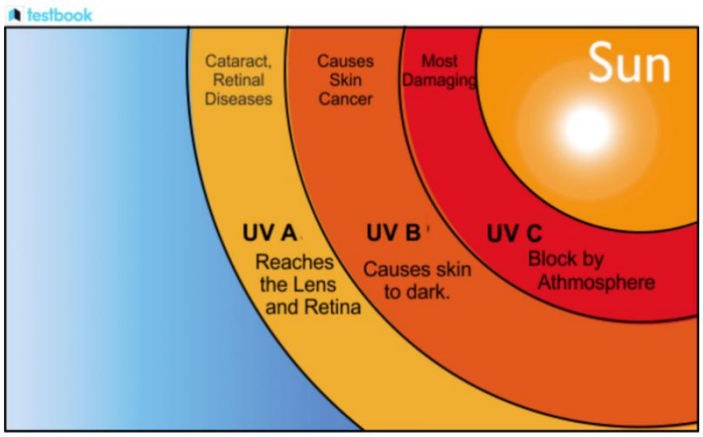


Figure 4 Different types of ultraviolet light [4]

In addition to this, there are two types of ultraviolet light, Deep Ultraviolet (DUV) and Extreme Ultraviolet (EUV). The wavelength of DUV is generally between 200-300 nm, while EUV has an

extremely high frequency, extremely short wavelength and extremely high energy. These two types of ultraviolet light are generally found in the industrial sector.

- **Application of ultraviolet light**

UVA is commonly used in tanning beds, blacklight applications, and phototherapy for skin conditions like psoriasis. It also plays a role in curing inks and resins in industrial processes due to its ability to initiate photochemical reactions. UVB is essential in medical treatments, particularly in vitamin D synthesis and controlled phototherapy for skin diseases. However, overexposure can cause sunburn and skin cancer, necessitating protective measures. UVC, with strong germicidal properties, is extensively used for sterilization in hospitals, water purification systems, and air disinfection, effectively inactivating bacteria and viruses [5].

DUV is crucial in semiconductor manufacturing, particularly in photolithography for producing microchips with high precision. Its shorter wavelength allows for finer circuit patterns in electronics. Meanwhile, EUV is revolutionizing advanced chip fabrication, enabling the production of smaller and more powerful semiconductors. EUV lithography is key to developing next-generation processors for computers and smartphones [6].

2) Ultraviolet Photodetectors

Since ultraviolet light is an invisible light wave, people cannot observe the presence of ultraviolet light in the environment with their eyes in daily life. As a result, there is an ultraviolet photodetector, which is a photoelectric conversion device based on wide bandgap semiconductor materials (such as ZnO, GaN, SiC, etc.), which can efficiently respond to optical radiation in the ultraviolet band and convert it into electrical signals. Put simply, the device "sees" ultraviolet light and converts it into a measurable current or voltage signal through the photoelectric effect.

- **Applications**

Ultraviolet light detectors are used in many fields. In terms of environmental monitoring, it is used for ozone layer hole detection and real-time monitoring of ultraviolet index to provide accurate sun protection guidance to the public; In the field of military security, this technology can sensitively detect missile tail flames and fire hazards, and is applied to sun-blind ultraviolet confidential communication with strong anti-interference ability; In terms of healthcare, UV detectors are not only used for radiation dose monitoring in skin cancer treatment, but also show value in cutting-edge fields such as cell biology research and DNA analysis. In industrial applications, the device provides reliable guarantee for key processes such as arc discharge detection and UV curing process control. In addition, in emerging fields such as aerospace remote sensing, criminal investigation, such as fingerprint fluorescence detection, and smart wearable devices, ultraviolet photodetectors have also shown broad application prospects. With the advancement of materials science and microfabrication technology, its performance continues to improve, and the application boundaries continue to expand [7].

- **Materials**

The development of semiconductor materials began in the mid-20th century, marked by the purification and preparation of single crystals of silicon (Si) and germanium (Ge). In the 1950s, silicon became mainstream with its excellent stability, abundant reserves and mature oxidation process, laying the foundation for integrated circuits. After the 1970s, III-V compounds (such as GaAs, InP) were used in the fields of communication and optoelectronics due to their high-frequency and optoelectronic properties. Since the beginning of the 21st century, wide bandgap semiconductors (such as ZnO, SiC and GaN) have promoted the development of power electronics and 5G technology due to their high voltage and high temperature resistance. At the same time, two-dimensional materials (such as

graphene), oxide semiconductors (such as IGZO), and quantum dot materials provide new directions for flexible electronics, display technology, and quantum computing. In the future, semiconductor materials will continue to evolve towards higher performance, lower power consumption, and new functionalization.

- **Structure and type**

The structure of ultraviolet photodetectors is mainly divided into photoconductive type, photovoltaic type, avalanche photodiode (APD), phototransistor and new heterojunction structure. The photoconductive detector uses a simple two-terminal electrode design to excite the carriers of semiconductor materials (such as ZnO and GaN) with ultraviolet light, causing a change in conductivity, but it needs to be externally biased and has a slow response. Photovoltaic detectors include PN junctions/PIN junctions (e.g., SiC, GaN-based) and Schottky junctions (e.g., Pt/ZnO), which separate photogenerated carriers through a built-in electric field, eliminating the need for external bias and fast response. The avalanche photodiode (APD) adopts an absorber layer and a multiplier layer structure to achieve high gain detection at high reverse bias voltage, which is suitable for low-light environments. Phototransistors, on the other hand, combine light detection and signal amplification through a three-terminal design (e.g., ZnO-based). In addition, novel heterojunction structures (e.g., ZnO/GaN, organic/inorganic hybridization) can further improve their performance through band manipulation to meet the needs of high sensitivity, fast response, or flexible integration. Different structures have their own advantages and can be selected according to specific application scenarios [8].

3) Experimental Significance

UV photodetectors are pivotal elements in UV sensing systems, with performance enhancement cantered on attaining minimal noise, elevated responsivity, and rapid reaction. Consequently, designing novel ultraviolet detectors featuring swift response, heightened sensitivity, and economical production has emerged as a primary focus in contemporary research.

Among potential materials, zinc oxide (ZnO) demonstrates notable benefits owing to its distinctive attributes: a room-temperature bandgap (3.37 eV) renders it optimal for solar-blind UV detection, an exciton binding energy as high as 60 meV guarantees superior quantum yield, alongside favourable chemical stability and crystalline quality. In contrast to alternative wide-bandgap semiconductors like SiC and GaN, ZnO also presents clear cost efficiencies. These traits position ZnO as a highly promising candidate for ultraviolet detection applications.

ZnO has emerged as a critical material for UV detector studies due to its near-perfect band alignment, outstanding optoelectronic performance, and robust physicochemical stability, with substantial advancements achieved in device development. The material not only possesses high charge carrier mobility and exceptional radiation hardness but also delivers reliable photo-response behaviour. Notably, the vertical architecture of ZnO-based detectors can efficiently reduce electrode separation by minimizing the photosensitive layer thickness, thereby boosting device efficacy. Nevertheless, further augmenting response velocity, improving sensitivity, and lowering fabrication expenses remain central challenges in current ZnO ultraviolet detector research.

4) Experimental Objectives

In this experiment, a Ni/n-ZnO/Ni horizontal structure ultraviolet photodetector was prepared. A single layer of ZnO film was first sputtered on a sapphire substrate by magnetron sputtering. Then, the interdigital electrode was etched by ultraviolet lithography, and the specific steps were spin coating photoresist, pre-baking, UV lithography, development, and post-baking. The Ni electrode was sputtered under vacuum by magnetron sputtering. The completion of the degumming step is considered to have

completed the preparation of the interdigitated electrode. Then, the photoelectric performance of the Ni/n-ZnO/Ni horizontal structure ultraviolet photodetector was tested and analysed, and the voltammetry characteristic curve, the responsivity at different wavelengths, the switching time of the device under five irradiation cycles, and the comparison of the optical response time under different bias voltages were studied. Finally, the experimental results were analysed to see if they were in line with expectations.

The expectation is that the current-voltage (I-V) curves of the Ni/n-ZnO/Ni horizontally structured photodetector should exhibit symmetrical and near-linear I-V characteristics under dark and 360 nm light irradiation, while exhibiting the same photo response characteristics at positive and negative biases. At the same time, under the dark and light irradiation of different wavelengths from 320nm to 400nm, the scanning voltage range is -3~3V, and a symmetrical and almost linear I-V curve is also observed, which is in line with the ultraviolet light response characteristics. By analysing the responsivity at different wavelengths, the cut-off wavelength of the device can be calculated, and whether the cut-off wavelength of the ZnO material is within the normal range (about 360nm-380nm). Applying light to the device for 5 s and removing light for 5 seconds for five cycles should increase as the light is irradiated and then decrease as the light is turned off. Comparing the optical response times (0.5 V, 1 V, 1.5 V, and 2 V) at different bias voltages, it is expected that larger voltages can cause large changes in photocurrent.

II. Experimental Principles

Thanks to Einstein's clarification of the internal mechanisms of the photoelectric effect that subsequent scientists were able to develop semiconductor devices that convert ultraviolet radiation into electrical signals.

In 1905, Einstein successfully explained the phenomenon of electron release from the surface of metals when the frequency of light exceeds a certain threshold, using the photon hypothesis in his study of the photoelectric effect. Experiments showed that the energy required to release electrons is proportional to the energy of the photons, satisfying the equation $E = hv$ where E is the photon energy, h is Planck's constant, and v is the frequency of the photon. The energy that a photon carries has a proportional relationship with the frequency [9].

1) Working Principle and Impact Factors

Semiconductor photoconductivity is the cornerstone of UV photodetection. It is based on the principle that when a semiconductor is exposed to light having a photon energy (hv) greater than its bandgap energy (Eg), electrons in the valence band are promoted to the conduction band. This process increases the number of free-charge carriers and, consequently, the electrical conductivity of the material [10].

For photoexcitation to occur, the energy of the incident photon must satisfy the condition:

$$h\nu \geq h\nu_0 = Eg \quad (1)$$

where h is the Planck constant and ν represents the frequency of the incident light. This photogeneration process creates electron-hole pairs that modify charge carrier density, thereby altering electrical conductivity and producing measurable photocurrent [11].

In the dark, the conductivity (σ) of the material is determined by the equilibrium carrier concentrations (n_0 and p_0 for electrons and holes, respectively) and their mobilities (μ_n and μ_p):

$$\sigma = q(n_0\mu_n + p_0\mu_p) \quad (2)$$

Under illumination, the generation of photo-induced carriers alters the carrier concentration. Let Δn and Δp represent the non-equilibrium carriers created under light exposure. Upon excitation to the conduction band, these carriers may carry excess energy beyond that of the thermal equilibrium carriers. However, due to collisions with the lattice, they quickly lose excess energy in the form of phonons and return to equilibrium. It is essential to note that in the process of photoconductance, the mobility of photogenerated electrons can be considered equivalent to that of the thermally balanced carriers [12].

The photogeneration of additional carriers creates a non-equilibrium concentration, producing an enhanced conductivity (σ):

$$\Delta\sigma = q(\Delta n\mu_n + \Delta p\mu_p) \quad (3)$$

The change in conductivity ($\Delta\sigma$) attributable to photoexcitation is critical in determining the detector's sensitivity [13]. In most high-quality semiconductors, impurity induced photoconductivity contributes the least to the intrinsic photoconductivity effect. Compared with the intrinsic photoconductivity of the device, impurity photoconductivity can be ignored in the detection process because the number of impurity atoms is significantly smaller than the number of crystals themselves.

In conclusion, ultraviolet photodetectors leverage the fundamental principles of semiconductor photoconductivity to convert incident UV radiation into an electrical signal. The basic mechanism involves several key processes: photon absorption, electron–hole pair generation, carrier transport and signal collection.

A simplified schematic of the charge generation and separation process in a typical UV photodetector is presented below (Fig. 5).

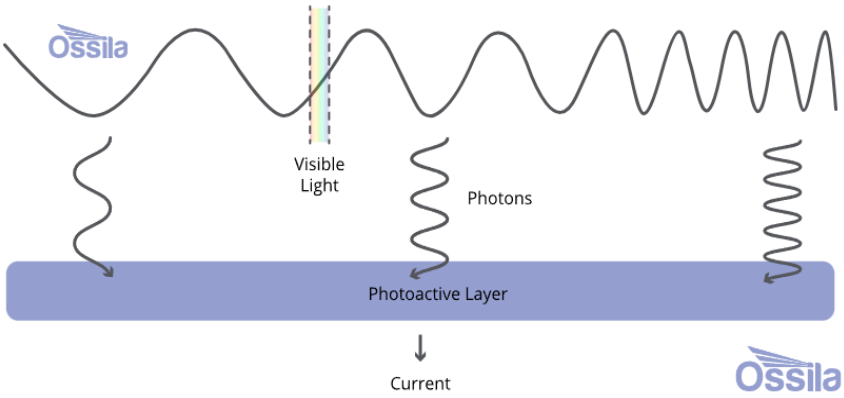


Figure 5 Working principle of Photodetectors [14]

III. Experimental Instruments and Materials

1) Instruments and Equipment

- **Spin coating machine**



Figure 6 Spin coating machine used in lab

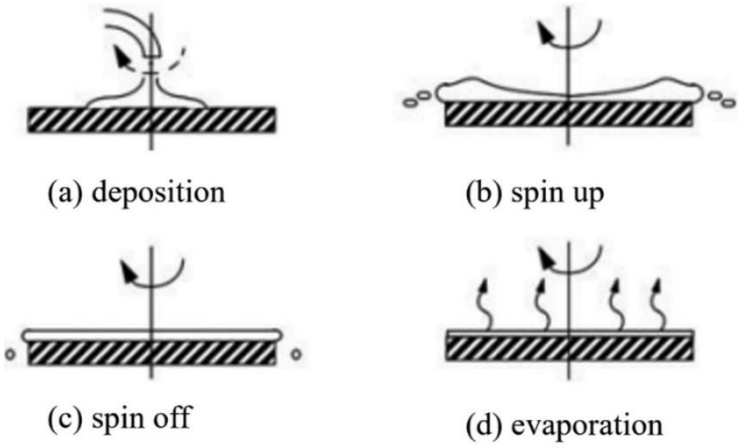


Figure 7 Principle of spin coating machine [15]

Spin coating, in past few decades, has been utilised in thin film manufacturing. The substrate is set to spin around a certain axis perpendicular to the coating area. In this process, the final film thickness as well as other properties are affected by the viscosity, drying rate, percent solids, surface tension and other natures of the coating, along with the arguments set for the machine [16]. Fig. 6 displays the different perspectives of the spin coating machine used in the experiment. As shown in Fig. 7, spin coating is divided into 4 steps [17]. To complete this procedure, spin coating machine is used to fix the sample by vacuum and then start spinning to spread the sol through the principle of centrifugal force.

In the real test procedure, totally 2 different revolution speeds were used. The former one, which was slower than the latter, was used to coat the sol in short time, while the latter one made the sample thinner. In this case, the sol is evenly coated on the substrate. For the evaporation of solvent, another apparatus called heater is used in the experiment.

- **Heater**

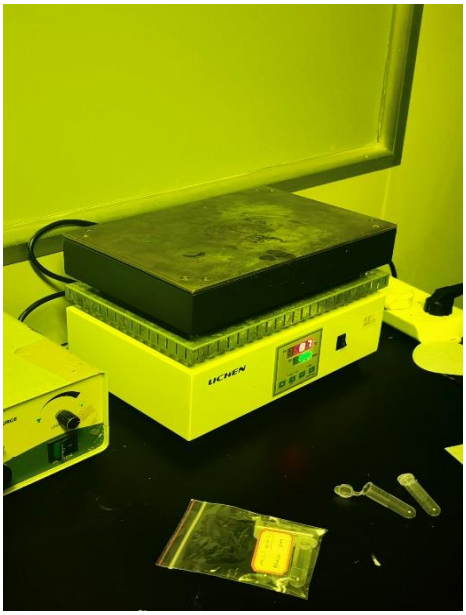


Figure 8 Heater

Fig. 8 shows the heater used in the experiment. The heater maintains a constant temperature to help the solvent to evaporate. The photoresist is reserved on the substrate after all the solvent are gone. In this case, the stability and completion of the sample can be ensured in the meantime of raise the work efficiency.

- **UV lithography machine**

Lithography is an effective method of nanoarchitecture development which is suitable to produce nanometre-sized patterns. With the assist of deposition and etching techniques, patterns with high-resolution are produced. Therefore, it is a popular technique that is often used to fabricate electronic microstructure devices. Lithography can be simply classified into two types: masked lithography and maskless lithography. Masked lithography, including photolithography, soft lithography and nano-print lithography, which allows patterns to be created under masks. However, arbitrary patterns are produced by maskless lithography technique, which includes *E*-beam lithography, scanning probe lithography and focused ion beam lithography. Precise patterned exposure is helpful to produce target micro patterns. Hence, lithography technique has high requirements on precision and is regarded as a sophisticated and costly technology. It also costs time and requires accurate sanitary controls of environment and operations [18][19].

A URE-2000/17 UV Lithography machine with an exposure wavelength of 365 nm was used in the experiment (Fig. 9). it consists of a compressor, a light, a mercury lamp, a mask plate and a monitor. The light is responsible for simple lighting work. A vacuum environment is created, which ensures the sample contact closely with the lens. Over the lens, the mask plate is placed. The mercury lamp produces beams of ultraviolet light, which go through the pattern on the mask plate and reach to the photoresist surface. The exposure under ultraviolet light breaks polymer chains in photoresist and decomposes those polymers into smaller ones, which have less molecular weights so that they are easier to be solved. With the assist of development process, along with the separate and leave of those smaller polymers, the pattern on the mask plate is displayed on the substrate.

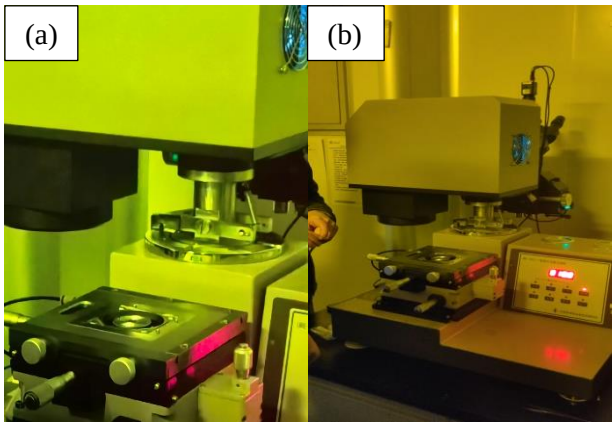


Figure 9 (a) Close shot of the machine; (b) UV Lithography machine

- **Microscope**

Microscope, as a commonly used equipment in laboratory, is mainly used to complete observation tasks. Microscope in this experiment is used to observe the sample in any steps of manufacturing (Fig. 10). Through careful observation, defects and completion are explicitly displayed on the monitor (Fig. 11), which provides the opportunities for timely feedback and adjustment to save research time.



Figure 10 Microscope in lab

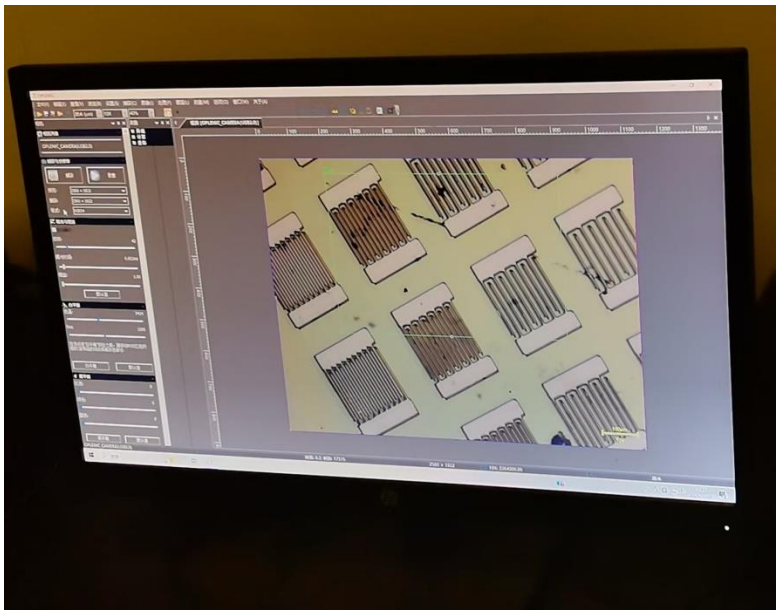


Figure 11 Monitor connected to the microscope

- **Magnetron sputtering system**

Thin film deposition is a popular technique of devices manufacturing which are often used in computers. Especially, thin film deposition plays a significant role in the manufacture of semiconductors. Semiconductor device fabrications, as an emerging industry, is mainly based on materials created by deposition technology. Thin film deposition can be classified in two ways: Chemical vapor deposition (CVD) and physical vapor deposition (PVD).

Sputtering is a PVD method, which is now the dominant technique in metal deposition. In this experiment, a magnetron sputtering system was used. In the last decade, magnetron sputtering has developed so fast that at now, it has acknowledged as the first choice for the deposition technique in various industrials. It is the rising number of needs of higher quality thin film in markets that promotes the high-speed development. In many cases, magnetron sputtered films perform better than other films deposited by other PVD processes. Meanwhile, they function same as "thicker films produced by other surface coating techniques". Therefore, magnetron sputtering now has important influence on different application fields especially for coating [20].

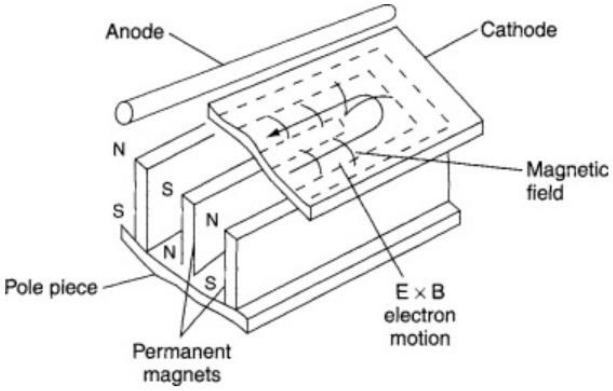


Figure 12 Sketch map of magnetron sputtering [21]

In magnetron sputtering, the combination of electric and magnetic field can rise the ionization rate of the electrons. Fig. 12 draws the sketch map of the theoretical mechanism of magnetron sputtering. A magnetic field B is created by magnets. The electric field E exists between the cathode and the anode. Under the join force of the Coulomb force driven by E and the Lorentz force driven by B, the moving trajectory of electrons is determined. The drift velocity vector v can be expressed by $E \times H$, where × represents a vector cross product. Furthermore, with the impact of the directional velocity, the moving trajectory of electrons, "a helical path around the magnetic field lines", is finally created. In this case, the electron path lengths are extended, which rise the residence time of electrons in the working gas and the colliding chances of particles, to increase the ionization efficiency of the gas. Hence, a higher sputter deposition rate is achieved [21].

The system used in the experiment is composed of a console, a thin film deposition system, a hydrocooling equipment and a set of high-pressure containers (Fig. 13). The console is mainly used as an integrated multistep power supply switch and a related parameter monitor, which can, for example, detect the flow rate. The thin film deposition system consists of a sputtering vacuum chamber, a magnetron sputtering target, a self-rotating and a heatable substrate stage.

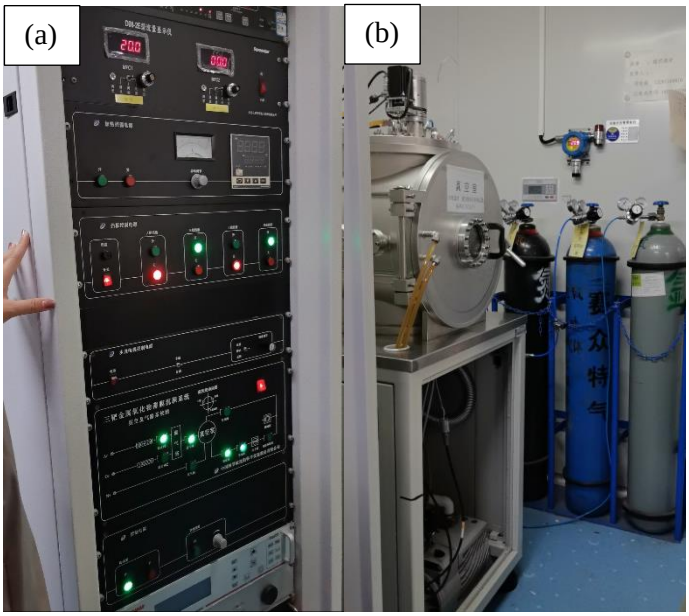


Figure 13 (a) Console in lab; (b) Film deposition system and air containers

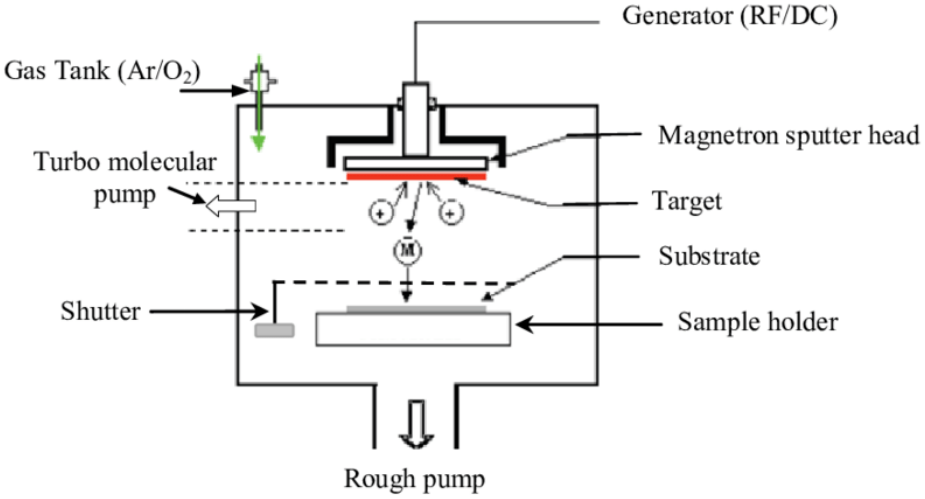


Figure 14 Structure of a thin film deposition system [22]

Fig. 14 shows the simplified structure of an often-used thin film deposition system. In the sputtering process, a vacuum environment is created by the air pump and molecular pump under the control of the console, which allows ions to speed up to sputter and reach the target. In the vacuum environment, proper gas is imported (inert gas such as Ar, sometimes with O_2 for oxides). Electrons collide with gas, which produces plenty of ions. Ions are driven by *E* to collide with the target, causing atoms to leave the target and land on the substrate.

- **Ultrasonic cleaning machine**

Fig. 15 shows the ultrasonic cleaner used in the experiment. When running, ultrasonic cleaner produce oscillated bubbles which impacts with sample [23]. Under this mechanism, the cleaner contacts with the sample closely and thoroughly, which facilitates the target gel to be removed successfully, and the interdigital electrode pattern to be retained.



Figure 15 Ultrasonic cleaning machine in lab

- **Light response testing system**

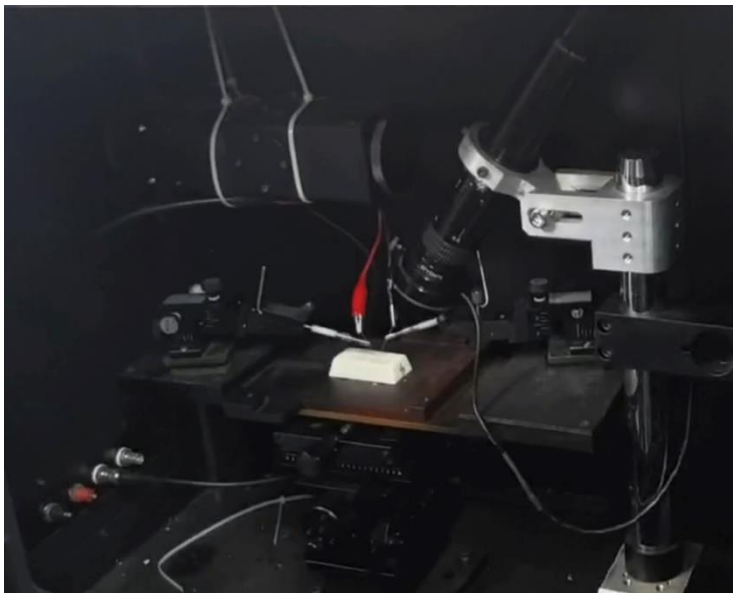


Figure 16 The insight of the darkroom

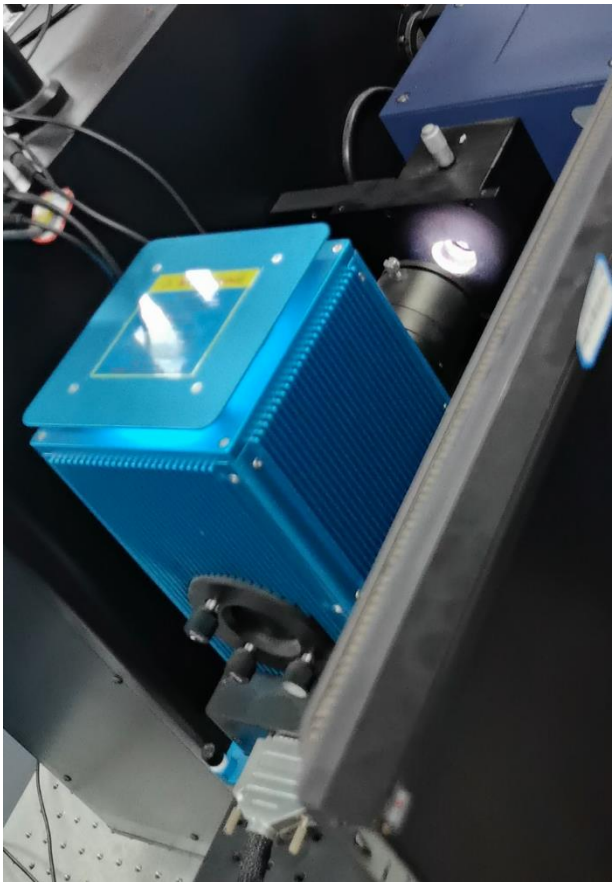


Figure 17 Light source and spectrometer

The light response testing system used in the experiment is a self-built system, including light source (deuterium lamp or xenon lamp), spectrometer, focusing mirror and other optical components, probe station, and semiconductor parameter tester (Fig. 16 and Fig. 17).

An optical spectrometer can separate lights with different wavelengths and is the most common type. They take light, separate it by wavelength and create a spectrum displaying the relative intensity of their wavelengths. For its detailed structure, optical spectrometer is composed of an entrance slit, a diffraction grating or prism, a detector, and routing optics. The entrance slit takes light in equipment, the lens routes light to the prism, The light is taken in by the entrance, routed the diffraction grating or prism by the lens or mirrors, and finally received by the detector. In this process, the prism separates light into various wavelength components. After that, the light intensity is recorded by the detector. Most optical spectrometers function within the range of the UV, visible, and infrared [24]. Optical spectrometers have a huge range of applications, including Agriculture, Biology, Earth and Planetary Sciences, Neuroscience, Physics and Engineering fields.

The semiconductor parameter tester is a sophisticated equipment that measures and tests semiconductor devices' performances, specifically the electrical characteristics. Semiconductor devices can be divided into two-terminal and three-terminal devices according to their different raw materials and manufacturing process. A semiconductor parameter tester can measure the I-V characteristics, C-V characteristics, breakdown characteristics, and other arguments. Engineers are allowed to use semiconductor parameter testers to examine the response of the devices and ensure their feasibilities. Therefore, semiconductor parameter testers become more important in the semiconductor industry [25].

In the experiment, by operating the system, along with the help of the light source and the optical spectrometer, lights with different wavelengths are produced separately and shot to the device. In this case, the electrical characteristics of the device are explicitly illustrated on the monitor.

2) Materials

- **Solid materials**

  Sapphire: Al_2O_3 single crystal. Sapphire is chosen as the substrate relying on its satisfied stability under heat and high mechanical strength, which allows material to grow and ensure the simplicity of cleaning.

  ZnO: a commonly used material for UV photodetectors. It has a bandgap width of 3.37eV at room temperature, which is classified as wide bandgap semiconductors. Apart from that, ZnO has good stability, easy crystallization, high quantum efficiency, high exciton binding energy (60 meV), and not easily oxidized. Furthermore, it is relatively cheaper.

  Ni: a typical conductive metal. It has remarkable ductility and a relatively higher melting point, allowing engineers to process. Ni is chosen as the sputtering target material relying on its corrosion resistance and high purity, which help to resist mixing of impurity. Its uniform structure ensures the reliability of the products.

- **Vapor materials**

  N_2: has stable chemical properties. It is used to dry and clean the vacuum room.

  O_2: used in the ZnO sputtering process. It ensures oxidation and can improve conductivity.

  Ar: an inert gas. It is the main gas used in the sputtering process. Ar does not react with targets and substrates, has suitable atom weight, and has low ionization energy. Hence, it can avoid impurity, raise sputtering efficiency and easily form plasma.

- **Liquid materials**

  Ethyl alcohol: C_2H_6O. It is often used as a cleaner due to its volatility.

  Acetone: $CH_3(CO)CH_3$. It is a polar solvent, whose O atoms in the molecules can form hydrogen bonds with the polar group in the photoresist. Therefore, acetone is used to lift off the photoresist.

  Deionized water: water without ion-type impurities. It is used as a cleaner so that impurities are isolated to the sample.

  Developer: an alkaline solution corresponding with the photoresist. It is used in the development to form the pattern after lithography.

- **Gel**

  Photoresist: composed of resin, sensitizer, solvent and additive. A positive photoresist is used in the experiment, where chemical reaction takes place after being exposed under UV light, which produces a type of acid that can easily be removed by the developer.

IV. Experimental Procedures

1) Device Structure Design Scheme and Optimization Process

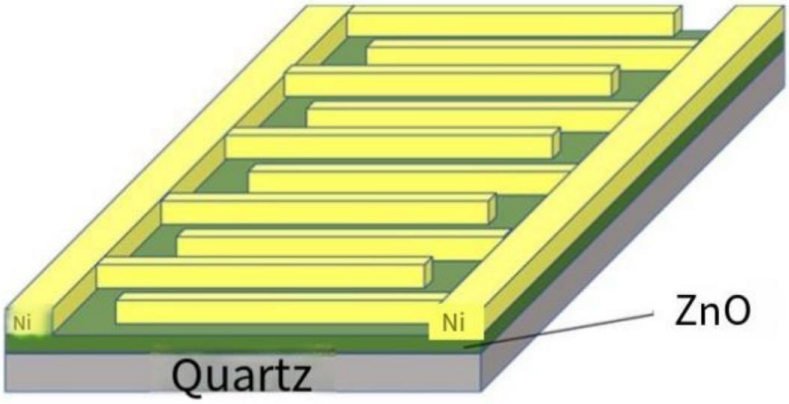


Figure 18 Basic structure of the MSM ultraviolet photodetector

The UV-photodetector designed in this experiment uses a metal-semiconductor-metal transverse structure shown in Fig. 18. This device contains a ZnO semiconductor film deposited on a sapphire substrate, with a pair of digital nickel electrodes on its top surface. This structure is easier to manufacture because of its simple structure. And has effective photoconductivity in the ultraviolet spectrum range (200–400 nm).

ZnO has wide band gap (~3.37 eV), high exciton binding energy (60 meV), and good thermal stability and chemical stability. It is an ideal material for UV detection, electrode material Ni has good adhesion, and the formation of near ohmic contact with ZnO, can achieve symmetric and linear I-V characteristics in lighting and dark conditions.

To achieve better optical response, an interfinger electrode pattern with a finger width of 10μm and a spacing of 20μm is designed. This geometry balances the resolution, carrier collection surface

area, and photogenerated charge extraction efficiency. The performance of the device is improved using a sapphire substrate with excellent lattice matching and good insulation properties with ZnO. To suppress leakage current, can also improve the signal-to-noise ratio.

2) Detailed Steps of the Manufacturing Process

The preparation process consists of the following major steps, which are substrate cleaning, deposition of ZnO film by RF magnetron sputtering, photoresist coating by spin coating, UV lithography, and Ni electrode deposition and then stripping, each step of which is performed in accordance with the corresponding standard operating procedures. This ensures the standardization and accuracy of the process and also ensures the reproducibility of the process. The detailed operations for each stage—including instrument setup, operational procedures, parameter configuration, and safety precautions—are elaborated as follows:

- **Substrate cleaning**

For sapphire substrate, it should be placed in acetone, ethanol and deionized water for ultrasonic cleaning, each ultrasonic cleaning time is 10 minutes, and then use nitrogen to dry it, the purpose of doing so is to ensure that organic pollutants, polar pollutants and particulate pollutants can be removed before film deposition.

- **ZnO thin-film deposition (RF sputtering)**

The cleaned substrate was loaded into a magnetron sputtering chamber. After achieving a base pressure below 5×10^{-4} Pa, argon and oxygen gases (the concentration ratio is 18:2) were introduced. The substrate was heated to 300°C, and an 80 W RF power was applied to a ZnO target. Sputtering was performed at 2 Pa for 130 minutes.

- **Photoresist coating**

After the ZnO film deposition, to determine the digital interelectrode pattern on the film using standard lithography techniques, the first step in lithography is to coat the ZnO/sapphire sample with a uniform positive photoresist, to achieve this goal, a rotating coating system is used. And in accordance with the rotary coating machine SOP to operate and photoresist processing. This process is carried out in a yellow light environment to prevent premature exposure of the photoresist, the specific operation is as follows:

Equipment start-up (Fig. 19): First turn on the **main power switch** of the spinning machine and then turn on the **vacuum pump** that provides suction for the spinning machine suction cup. Ensure that the rotating coater is placed on a level surface, and the inside of the rotating bowl is clean and clean. According to the requirements of standard operating procedures, the operator should stand while

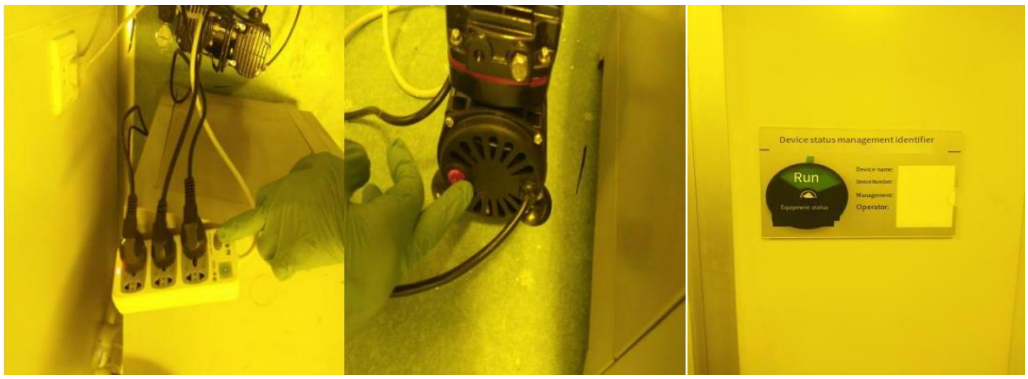


Figure 19 Spin coater's operation panel

operating the spinning machine. This is done in order to maintain the stability of the operation process, but also to ensure safety.

Program spin parameters (Fig. 20): First press the *Parameter Set/Modify* button on the controller panel and then use the interface to navigate the parameters with the up/down arrow keys to navigate parameters and plus/minus keys to adjust values and then enter the desired rotation configuration file. In this case, the two-step rotation is programmed, the **first step** is set to 600 RPM for 8 seconds, which is done to spread the resist evenly, the **second step** is set to 4000 RPM for 30 seconds, which is done to achieve the target thickness of the resist, according to the standard operating procedure SOP. To set the unused steps to 0 revolutions per minute. After entering all the values, press the "*Enter/Run*" key to save the program.



Figure 20 Power supply and vacuum pump

Load sample (Fig. 21): Line the spin bowl with a piece of cleanroom lint-free paper (this will catch excess resist spun off, simplifying cleanup) and then place the ZnO coated substrate in the centre of the rotating chuck, making sure to use tweezers to handle the substrate at the edge where the gloves are worn. It is also necessary to ensure that the vacuum groove of the chuck and the substrate are aligned, to ensure that the substrate can be held safely.

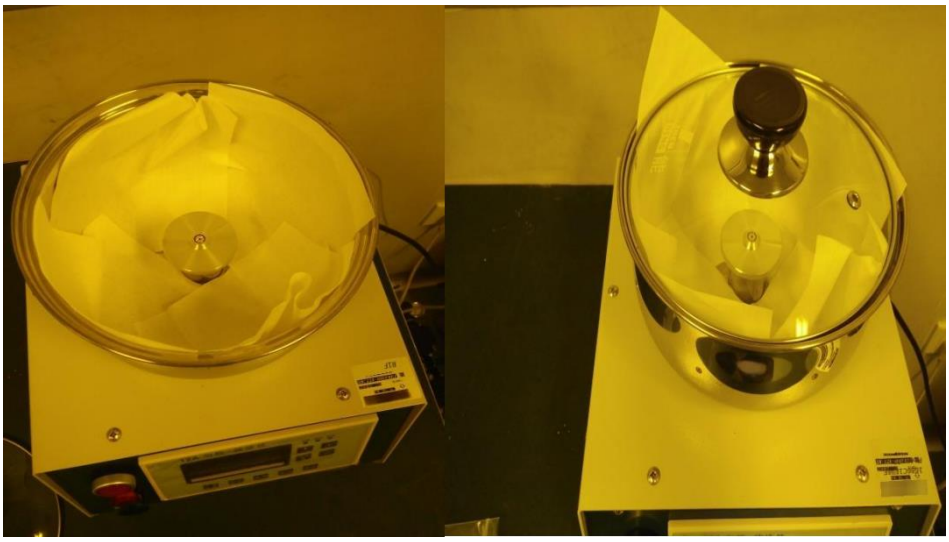


Figure 21 Spin bowl

Dispense photoresist: Use a disposable dropper or pipette to distribute enough positive photoresist on the surface of the substrate. In general, place 2 to 3 drops of photoresist in the middle of the sample, and the number of drops here may be about 0.1 ml, which depends on the size of the substrate. The amount of photoresist should be enough to cover the entire substrate. But do not use too much, otherwise it will cause the situation of thick edge beads.

Spin coating: Gently place the coater lid down to enclose the substrate (this minimizes turbulence and spreads solvent fumes into the exhaust). Next, press the **vacuum** button to allow the vacuum chuck to complete the engagement action, and then make sure that the substrate is firmly held. The specific way to confirm is to nudge the substrate, which should not slide under normal circumstances. After confirming the state of the substrate, then press the **Start/Spin** button, so that the rotation cycle of the program can be executed, the rotary machine will accelerate to 600 revolutions per minute after starting, and continue for 8 seconds, this 8 seconds time is used to let the resist to spread, after the resist diffusion is completed, the rotary machine will accelerate again to 4000 revolutions per minute. And for 30 seconds, with the help of these 30 seconds of rotation, the resist can be diluted to the thickness we need. It is important to note that during the entire spinning process, do not open the cover and do not approach the coating machine, because if the resist is not properly controlled, it is likely to spray out.

Soft bake (prebake): When the rotation cycle is completed, the lid is carefully removed, and the substrate is removed, when there is still wet solvent inside, and then the coated substrate is placed on a flat heating plate or in an oven set at 100°C for 25 minutes. This pre-baking operation can evaporate the solvent in the photoresist but also allow the adhesion between the photoresist and the ZnO film to be improved, if using a hotplate, set it to 100 °C and use a timer. After baking, let the substrate sit on a clean surface and slowly cool to room temperature.

- **UV photolithography**

After the photoresist is applied and baked, the next thing to do is to use ultraviolet lithography to carry out pattern processing work on the photoresist, we will use mask alignment or UV lithography system to transfer the digital interelectrode pattern from the mask to the resist. Here we use a UV exposure tool equipped with a 365 nm mercury lamp. In the process of operation, it is necessary to carry out in strict accordance with the standard operating process of UV lithography machine, which involves the correct order of a series of operations such as lamp ignition, alignment, exposure and closure, and the following details the specific operating procedures:

Lamp ignition and system setup (Fig. 22): Turn on the main **power switch** of the UV lithography system. Ensure that all necessary sub-systems are powered: the air compressor (for vacuum and stage flotation) is running, and the lamp power supply and microscope monitor are connected. Press the main power button (power strip or control panel switch) to supply power to the system. Next, start the **mercury lamp power supply** by pressing the red power button on the lamp control unit. Verify that the lamp's voltmeter shows a voltage (not zero) and the ammeter reads zero – this indicates the lamp power is on standby. Wait about 1 minute to allow the lamp's electrodes to warm slightly. Then press the mercury lamp **ignition (trigger)** button (often an orange button). A successful ignition will cause the voltmeter to drop to zero and the ammeter to show a current, indicating the lamp arc has struck. The mercury lamp will now start glowing; allow it to stabilize for a few minutes. *Caution:* The mercury lamp is very bright and emits UV radiation – the lamp housing should remain closed, and direct line-of-sight to the lamp should be avoided.



Figure 22 UV lithography machine's buttons

Control system activation (Fig. 23): Press the green **control power** button to turn on the mask aligner's control electronics and microscope system. This powers up the stage controls, vacuum controls, and alignment microscope camera/monitor.



Figure 23 Photomask

Mask loading (Fig. 24): Select the appropriate mask holder according to the size of the mask, the mask holder is sometimes called the adapter or drawer, slide the mask desk drawer out from the orthotic, and then place the mask on the mask table, so that the side with the chrome pattern is facing **down** towards the substrate, and align with the positioning pin or reference Angle. Typically, the mask should butt up against three alignment pins to ensure proper registration. After the mask is placed in a good position, press the **mask vacuum** switch on the control panel to activate the mask vacuum, so that the mask can be firmly fixed on the stage with the help of suction.

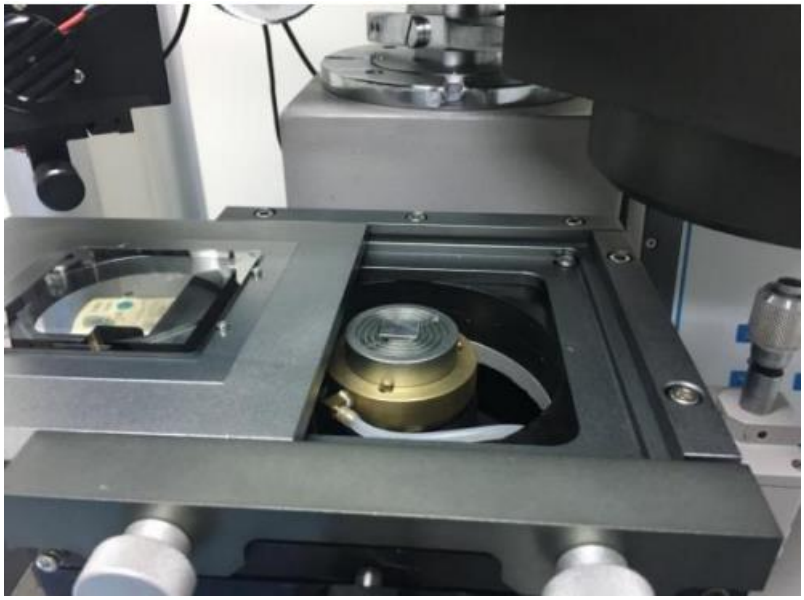


Figure 24 UV lithography machine's operation panel

Substrate loading: Before processing the photoresist coating sample, **turn off the white light, then turn on the UV-safe yellow light** in the calibration chamber, if the calibration chamber is not already in yellow light mode, to prevent any accidental exposure to the photoresist, then turn on the sample grade chuck, if the sample grade chuck is not yet out, Just slide the substrate bracket out. After that, gently place the roasted resistance coating substrate in the centre of the sample chuck, the sample should be positioned in the centre of the chuck vacuum port as far as possible. Now turn the **substrate vacuum** on. There is usually a switch or button marked "chuck vacuum" in the control area, which is turned on to ensure that the sample can be fixed on the platform. And if it is separate mask drawer also together, slide back to the aligner. Tighten any stage lock or fixture according to the actual needs, so that the stage can be fixed in the proper position.

Alignment microscopy (Fig. 25): First turn on the **alignment illumination** and also turn on the **video monitor** that shows the microscope view, which may involve rotating the microscope turret over the sample, then switch the lighting switch, and then adjust the microscope focus knob. Until the mask is clearly focused on the display. At this stage, you should be able to see the pattern of the mask and the substrate underneath it, especially if the resist or substrate has something like an alignment mark.

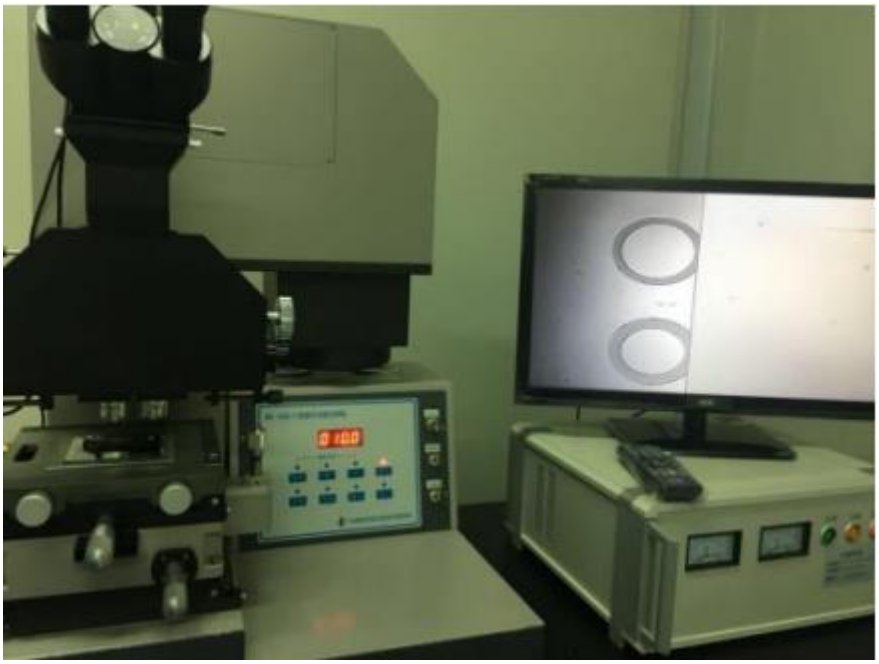


Figure 25 Alignment illumination

Contact and alignment: Rely on turning the **Z-axis up knob**, or pressing the button to control the table up, slowly and gently lift the substrate table, and then Continue until you feel a slight resistance or see that the substrate has contacted the mask (in a contact aligner, a slight Newton's rings pattern might appear, indicating contact). After the alignment is complete, the air float should be turned off so that the table can be locked, and if necessary, the table can be raised slightly to restore the close contact between the mask and the substrate.

Exposure setting: Set the exposure parameters on the control panel, press the *exposure time set* button, after pressing the button, enter the exposure duration we need. Use the controls (arrow or +/- keys) to adjust the time; In this experiment, the selected exposure time is **30 seconds**, and finally, press the setting button to confirm the setting until the display or indicator shows that the time has been registered.

UV exposure: Take protective measures to ensure that no one is observing the machine or exposed area without protection, if the UV lamp housing or mirror assembly is not fixed in the proper position, then swing them to the mask and substrate, and then press the **Expose** start button to begin UV exposure. The ultraviolet light emitted by the mercury lamp will pass through the mask and project the pattern onto the photoresist. After the exposure, you should **walk away immediately and cover your face** at the same time, to avoid any direct or reflected ultraviolet damage. The exposure time should be set, which is 30 seconds. The system will automatically close the shutter of the light. Finally, the Z-axis knob is used to lower the substrate level so that the mask can be separated from the substrate.

Unload sample and mask: First turn off the mask vacuum, do so to release the mask to avoid the mask sticking to the resist when you pull the stage out, then slide out the stage drawer, then carefully remove the mask, and put the removed mask into a secure container. The substrate is then removed from the chuck by turning off the substrate vacuum and lifting the sample. The sample can be operated

with tweezers on its edge while lifting the sample. The sample is covered immediately after removing the substrate. This is done to protect the resist pattern from light prior to development, and to safely turn off the mercury lamp if the exposure is no longer required. It should be noted here that the mercury lamp generally needs to be cooled before it can be reignited, and many systems recommend that if another exposure is about to take place, or if the mercury lamp needs to be turned off, then the lamp should be turned off in accordance with the provisions of the SOP.

Precautions:

Never look directly at the UV light or the illuminated mask with the naked eye. Always wear UV-blocking safety goggles when working near an operating UV source. Even the reflected UV from the wafer or mask can cause eye damage.

Ensure that it is always kept in a **yellow light environment** from the beginning of the photoresist application until after development. If accidentally exposed to the ambient light, part of the resist will be exposed, and the pattern will be destroyed. When the sample depends on the normal lighting area, it is necessary to cover the sample with aluminium foil or lid.

The photomask is a valuable and delicate item; handle it by the edges or frame to avoid scratches on the pattern and always ensure the correct side (patterned side) is facing the substrate. Misorientation will result in a blurred or improper pattern transfer.

Making good contact between the mask and the substrate is particularly critical for high resolution characteristics, making sure that the substrate is flat, making sure that there are no particles on the mask or wafer that might create gaps, and if the mask and the wafer are not in proper contact, UV rays may diffract, which can cause problems with inaccurate patterns. Such as the appearance of shadows or blurred edges

- **Development**

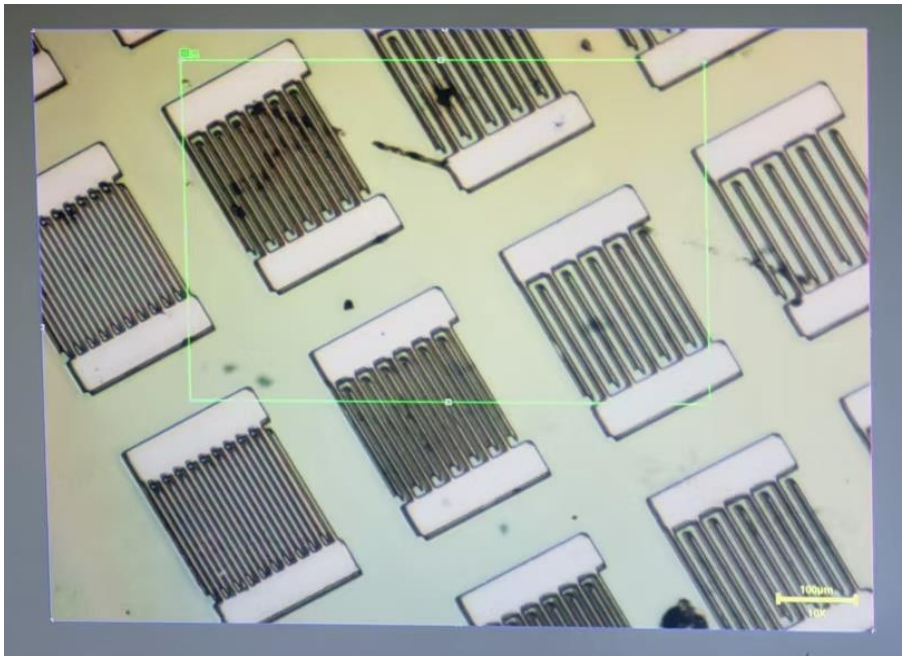


Figure 26 Developing the device

When the photoresist is exposed to ultraviolet light, it will be structurally modified, so that in a specific alkaline developer, it becomes more soluble than before. After the development operation, the UV exposed areas of the photoresist will be selectively dissolved and removed. The parts that are not

exposed to UV light remain intact above the substrate. The final remaining photoresist can precisely define the digital interelectrode pattern, which corresponds to the opaque area of the mask (Fig. 26).

- **Ni electrode deposition and lift-off**

Nickel electrode deposition and stripping is such that when the resist mode has been completed, the resist mode is the opening part of the resist that should deposit nickel, and the next final manufacturing step is to deposit nickel metal, by depositing nickel metal to form digital interelectrode, and then the stripping operation is carried out. This stripping operation is used to remove excess metal and photoresist. In the deposition of nickel, the same magnetron sputtering system is used to deposit nickel this metal, in the process of deposition of nickel, follow is similar to the deposition of zinc oxide standard operating procedures, however, in the deposition of nickel when there are different parameters, and also increased post-processing steps, the following details about the specific steps of nickel sputtering and lift-off.

Pre-sputtering preparations: Before putting the pattern sample into the sputtering chamber, to ensure that the resist pattern is clean, has been completely dry, any solvent contained in the wash solution should be evaporated, otherwise it will pollute the vacuum environment, to use the smallest mask, here the remaining resist itself can be used as a mask, the developed sample mounted on the sputtering sample holder.

Chamber pump-down (Fig. 27): If the sputtering system has been vented after the ZnO deposition, then the chamber needs to be pumped again, to place the resistance pattern substrate on the bracket, and then insert it into the chamber, while the operation to carefully hold the edge with gloves, so as to avoid damage to the resistance. To ensure that the nickel target is mounted on the cathode, here we use a 2-inch nickel target, which is 99.99% pure, and then evacuate the chamber to the basic pressure, which is about 10^−4 Pa, to remove the residual air in the resist, which contains the solvent.

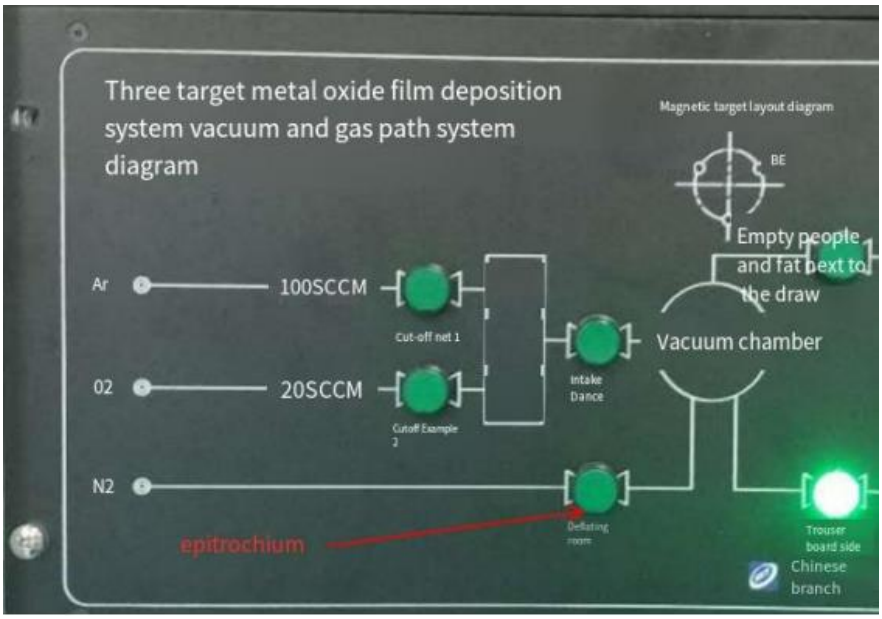


Figure 27 Vacuum and gases system

Argon gas and pressure setup (Fig. 28): It is necessary to introduce high-purity argon into the room, so that the room can reach the pressure of sputtering work, here to explain that when the deposition of metal is not added oxygen, because nickel will be oxidized if oxygen is added, and then the Ar flow rate is set to about 20 sccm, of course, it can be adjusted according to actual needs. Then

let the throttle pump work so that the pressure reaches **1 Pa**. Then wait a while for the pressure to settle down.

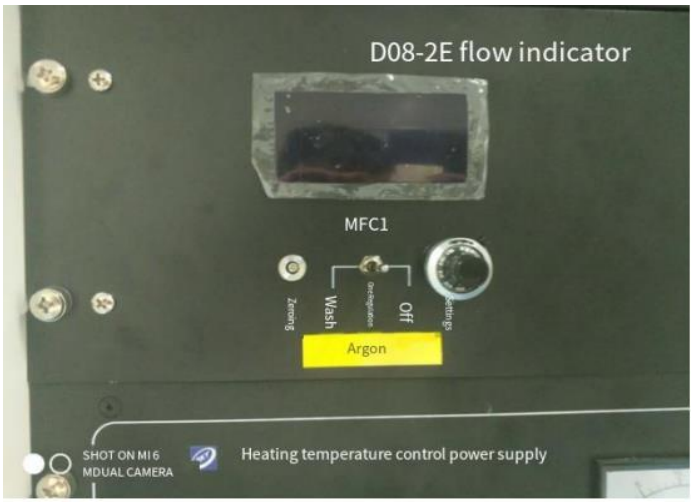


Figure 28 Argon flow controller and indicator

Plasma ignition for Ni (Fig. 29): Since nickel is a conductive metal, the DC sputtering power supply is used, the DC power supply is turned on first, and then slowly increase the power, until the required setting value is added, in terms of expansion, for Ni is to be added to **60 W**, when the power reaches 60w this setting value, let the plasma hit the Ni target, Note here that the plasma should be ignited in a characteristic glowing state. Then you have to make sure that the plasma looks stable and covers the target evenly.



Figure 29 Plasma ignition and shutters controller

Ni film deposition: When the plasma reaches a stable state, the shutter is opened and the nickel deposition work begins, the substrate can be stored at room temperature, this is because it does not need

to be heated, if the temperature is too high, the photoresist mask will be damaged, and then the nickel sputtering is carried out in the scheduled time. Like in the process we were doing, it was sputtering for **10 minutes** at 60w DC. After this operation, a nickel film of about 100 nm thickness is formed in the exposed area of the substrate, and it should be noted here that the exact thickness is actually related to the deposition rate and time, the nickel will be covered in various places on the substrate that are not covered by the photoresist, will be covered on the top of the photoresist covered area, and will be directly covered on the ZnO in the open area.

Completion of sputtering: After 10 minutes, close the target shutter, then reduce the DC power to zero, then turn off the DC power, then turn off the argon valve, and then turn off the flow controller. Let the chamber purge a bit and cool if necessary. At this time, the sample may be kept near room temperature. After draining the chamber in the same controlled manner as before, the sample is removed, and it is found that the sample is now evenly coated with a thin nickel film, both in the pattern area and on the top of the resist.

Lift-off process: To remove the excess nickel and determine the electrode pattern, Place the Ni-coated sample into a glass container with **acetone** (enough to submerge the sample). And then gently ultrasound this glass container with the sample and the acetone for a few minutes. In this process, the acetone will dissolve the underlying photoresist, which will cause the top nickel film to lift up and break, when the ultrasound treatment is carried out for 2 to 5 minutes, the unwanted nickel should be separated from the substrate, and finally only in the area directly deposited on the ZnO, that is, the open area in the resist. Then examine the sample to make sure that all the photoresist and the nickel covering it have come off, rinse the device thoroughly with isopropyl alcohol and deionized water, and then dry it.

Precautions:

When handling the developed resist-patterned substrate for Ni sputtering, avoid any contact with the resist features. Even a slight touch can break or deform the tiny resist lines, which would lead to defects in the electrode pattern. Hold the substrate by the edges only.

To keep the sputtering time and power in a moderate state to avoid overheating of the substrate, this is because the photoresist mask can only withstand limited heat, if excessive heating, it may make the resist backflow or burned, destroy the pattern, but also make the emission more complicated, after practice found that the use of 60 W DC, 1 Pa, A 10-minute operation is mild enough for the process we have now, without the problem of excessive heating.

During the lift-off, **do not rush** the process. Allow the solvent to penetrate and dissolve the resist fully. Aggressive poking or trying to peel metal off can also peel off the desired metal. Proper lift-off will result in the unwanted metal simply floating away.

Perform lift-off in a glass dish with a cover, inside a fume hood, as acetone is volatile and flammable. Ultrasonic baths can cause acetone to heat and evaporate quickly, so ensure the container is covered to minimize vapor. Also, avoid inhaling fumes and wear gloves and goggles.

After the samples are lifted up, they are examined under a microscope, looking for metal fragments left between the finger seams, or any other particles that could cause electrical short circuits or affect the performance of the equipment, and if they are found, additional cleaning steps may be required. Such as conducting another ultrasound examination or carefully removing them with a fine needle under a microscope.

Properly dispose of the used acetone and resist waste. It is hazardous waste and must be collected in an appropriate container per lab safety rules.

3) Circuit Connection and Performance Testing Steps

The performance of the fabricated Ni/ZnO/Ni MSM UV photodetector was evaluated using a laboratory-based photoelectric testing platform shown in Fig. 30. The system has a xenon lamp, which is used as an ultraviolet light source, a monochromator, which is used for wavelength selection, and a micro-positioning probe station, which is used for contact devices. And a semiconductor parameter analyser (SPA) for electrical characterization.

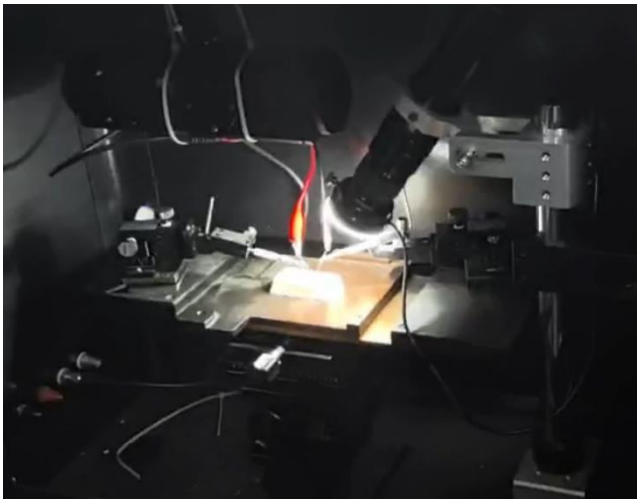


Figure 30 Photoelectric testing platform

- **Circuit setup and optical illumination**

The sample is mounted on a vacuum suction cup under the probe station. Two tungsten probes are carefully arranged so that they have access to the Ni electrodes. One electrode is connected to the positive electrode of the SPA, and the other electrode is grounded. The xenon lamp emits a wide band of ultraviolet light, which is directed by the monochromator to produce a monochromatic light. Next, the beam is aligned and focused to the effective region of the detector, the wavelength will change in the range of 240 ~ 400 nm, and then the spectral response is analyzed.

- **I-V characterization**

The SPA will be configured as a voltage scan mode, under dark conditions and illuminated conditions to record the current-voltage curve. This device exhibits approximately linear symmetric I-V characteristics, which indicates that it is ohmic contact.

- **Spectral responsiveness measurement:**

Responsivity was calculated using the formula

$$R = I_ph(P \times A) \quad (4)$$

where I_ph is the photocurrent, P is the incident power density, and A is the effective area. The measured wavelength is from 240 nm to 400 nm.

- **Temporal response:**

A mechanical chopper is used to chop the 360 nm beam to evaluate the time response of the detector. The current response is recorded at different bias voltages (1 V, 1.5 V, 2 V), and the device shows relatively fast rise and fall times, because the carrier mobility increases, and the response speed is also improved at higher voltages due to increased carrier mobility.

Precautions:

Ensure stable and consistent illumination power during measurement to avoid drift in photocurrent values.

Perform dark measurements in a light-sealed environment to prevent background exposure.

Avoid electrostatic discharge (ESD) when probing the device by grounding the operator and using ESD-safe tools.

Allow adequate rest time between measurements to minimize thermal effects caused by prolonged UV exposure.

V. Experimental Results and Analysis

1) Photodetection Performance results of Ni/n-ZnO/Ni horizontal structure ultraviolet detector under an Ar / O_2 Ratio of 18: 2

The laboratory photo response testing system was used to evaluate the optoelectronic performance of the Ni/n-ZnO/Ni horizontal structure ultraviolet detector. All figures were plotted using Origin software, including the current-voltage characteristics under dark and various wavelength illumination conditions, the responsivity at different wavelengths, the switching behaviour of the device over five light cycles, the comparison of light response under different bias voltages, and the response time.

- **Current-Voltage (I-V) characteristic curve**

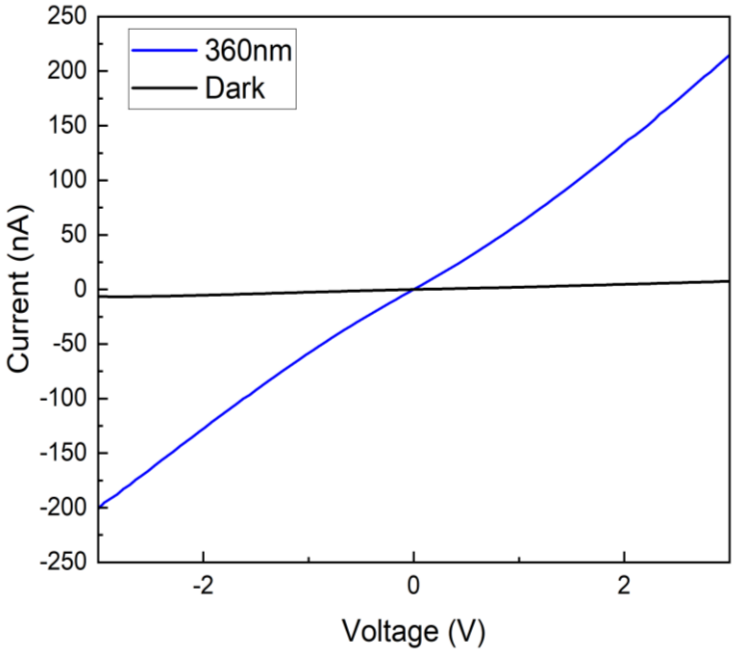


Figure 31 I-V characteristic curves of Ni/n-ZnO/Ni horizontal structure ultraviolet detector in the dark and under 360nm illumination conditions

Fig. 31 illustrates that Ni/n-ZnO/Ni horizontal structure ultraviolet detector exhibits symmetrical and nearly linear current-voltage (I-V) characteristics under both dark and illuminated conditions, indicating the formation of an ohmic metal-semiconductor contact. The type of contact is characterized by the seamless transfer of majority carriers between two materials, ensuring that current flow remains

unrestricted [26]. It is essential for the optimal performance of photodetectors. Additionally, the figure reveals that the Ni/n-ZnO/Ni horizontal structure ultraviolet detector exhibits identical photo response characteristics under both positive and negative bias. Moreover, the current detected under dark conditions is referred to as dark current, which mainly arises from thermally generated carriers and the saturation current of the semiconductor junction. Dark current increases with temperature and introduces additional background noise, thereby reducing the signal-to-noise ratio (SNR) and potentially hindering the detection of weak optical signals [27].

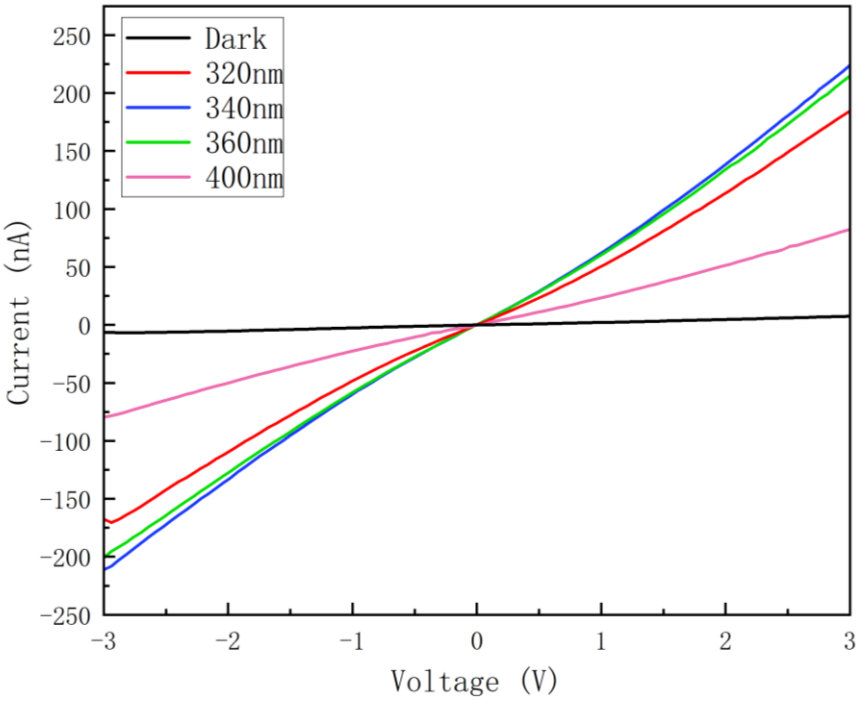


Figure 32 I-V characteristic curves of the Ni/n-ZnO/Ni horizontal structure ultraviolet detector in darkness and under illumination with different wavelengths from 320 nm to 400 nm

Fig. 32 presents the detailed photo response characteristics, primarily for various UV-A wavelengths ranging from 320 nm to 400 nm. Under both dark conditions and illumination at different wavelengths within this range, the scanning voltage spanned from -3 V to 3 V. The observed symmetrical and nearly linear I-V curves further confirm the UV light response characteristics of the Ni/n-ZnO/Ni horizontal structure ultraviolet detector.

- **Responsivity characteristics**

Responsivity characterizes how efficiently a photodetector converts incident optical power into an electrical signal, typically expressed in units of amperes per watt (A/W) [28].

The calculation formula is

$$R = \frac{I_p - I_d}{P_0} \tag{5}$$

where R is the responsivity, I_p is the photocurrent, I_d is the dark current, and P_0 is the optical power. According to Fig. 33, the responsivity of the experiment is high in the 320–360 nm band, with a maximum value of about 0.012 A/W. The responsivity gradually decreases with increasing

wavelength, drops rapidly after 400 nm, and is almost unresponsive in the 450–500 nm band. It is predicted that the cut-off wavelength of the device is approximately 440nm.

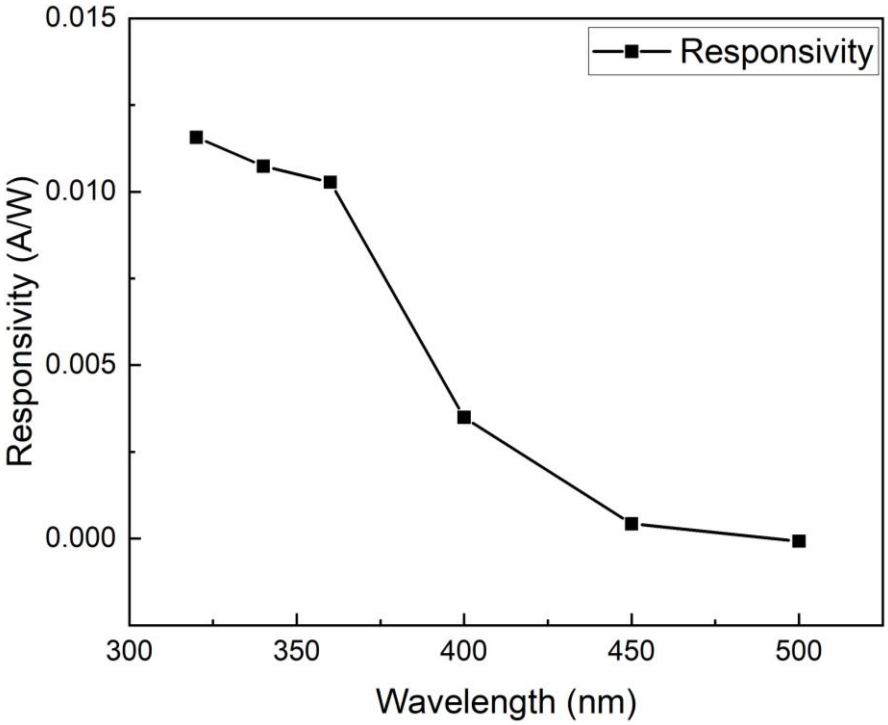


Figure 33 Responsivity curve of the Ni/n-ZnO/Ni horizontal structure ultraviolet detector

Theoretically, according to the photoelectric effect, photons can be absorbed by charge carriers and generate photocurrent only when the energy of the incident light exceeds the bandgap of the ZnO material. This enables electrons to transition from the valence band to the conduction band. ZnO is a wide bandgap semiconductor with a bandgap of approximately 3.3 eV, which corresponds to a wavelength of around 375 nm. Therefore, only ultraviolet light with wavelengths shorter than 375 nm can effectively excite electron transitions and contribute to photocurrent generation in ZnO-based photodetectors. This result deviates slightly from the responsivity curve we obtained from the measurement of the Ni/n-ZnO/Ni horizontal structure ultraviolet detector.

- **The temporal response characteristic**

The time-dependent photo response of the Ni/n-ZnO/Ni horizontal structure ultraviolet detector was systematically investigated under 360 nm illumination. Five cycles of applying illumination for 5 s and removing illumination for 5 s were performed at different bias voltages.

Fig. 34 shows the time-dependent current response under a forward bias of 2 V. Under the condition, the current increases upon illumination and decreases when the light is turned off. Fig. 35 shows the time-dependent photo response under 360 nm illumination at four different bias voltages: 0.5 V, 1 V, 1.5 V, and 2 V. For each of the four positive bias voltages, two on-off cycles were recorded, with each light-on and light-off period lasting 5 seconds. It can be observed from the figure that higher bias voltages facilitate the transport of photogenerated charge carriers, leading to more significant changes in photocurrent. Specifically, the photocurrent response at a bias of 2 V is approximately 1.5 times greater than that at 1.5 V, and about six times greater than those at 1 V and 0.5 V. Therefore,

under illumination with the same wavelength, the influence of higher bias voltages on the device's photo response are considerable and should not be overlooked.

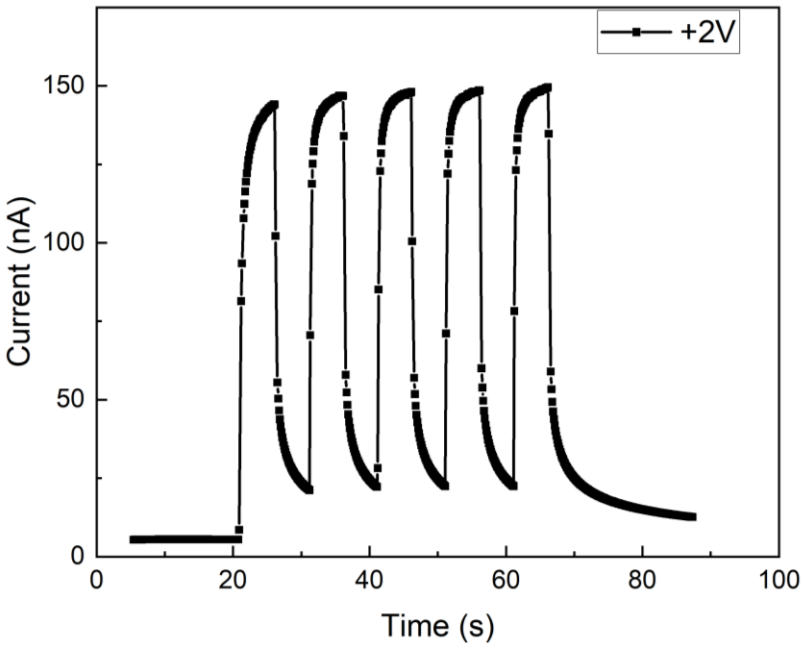


Figure 34 Temporal response of the Ni/n-ZnO/Ni horizontal structure ultraviolet detector under different bias and 360 nm illumination

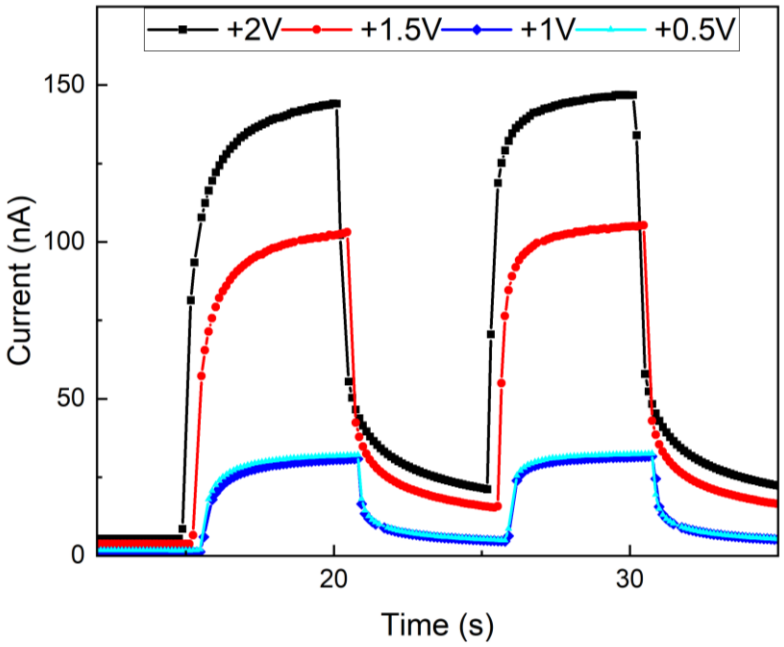


Figure 35 Temporal response of the Ni/n-ZnO/Ni horizontal structure ultraviolet detector under 2V bias and 360 nm illumination

The photo response curve can be fitted using a biexponential equation [29].

$$I(t) = I_0 + A_1 e^{\left(-\frac{t}{\tau_1}\right)} + A_2 e^{\left(-\frac{t}{\tau_2}\right)} \quad (6)$$

In the above formulae, I_0 is the initial current, A_1, A_2 are amplitude coefficients, τ_1 and τ_2 are time constants. Take the maximum bias voltage of 2V as an example, it can be found from its fit curve data shown in Fig. 34 that the time constant of the rapid rise and decay process is approximately $\tau_{r1} = 0.43s$ and $\tau_{d1} = 0.20s$, while the time constant of the slow rise and decay process is about $\tau_{r2} =$

$5.31s\ and\ \tau_{d2} = 2.00s$. Fig. 35 presents both the rising fit curve that is green and the purple fit curve of the decline.

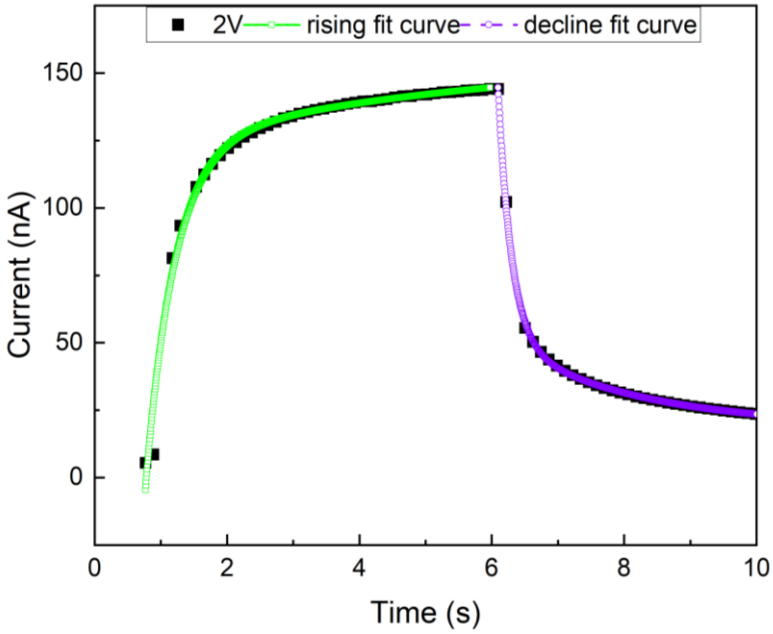


Figure 36 Temporal response of the Ni/n-ZnO/Ni horizontal structure ultraviolet detector under +2V bias and 360 nm illumination and fitting curve

Model	ExpDec2
Equation	y = A1*exp(-x/t1) + A2*
Plot	[A360nm2V]"360nm 2V"!B"
y0	157.08968+/-40.14443
A1	-744.87642+/-118.0697
t1	0.43701+/-0.05668
A2	-38.68233+/-21.02889
t2	5.31392+/-12.17456
Number of Points	42
Degrees of Freedom	37
Reduced Chi-Sqr	13.95832
Residual Sum of Squres	516.45793
R-Square (COD)	0.98626
Adj. R-Square	0.98478
Fit Status	成功(100)

Model	ExpDec2
Equation	y = A1*exp(-x/t1) + A2*
Plot	[A360nm2V2]"360nm 2V"!B
y0	18.82761+/-0.53673
A1	680.22033+/-135.60593
t1	2.00494+/-0.12339
A2	1.46724E15+/-9.67929E14
t2	0.20066+/-0.0044
Number of Points	42
Degrees of Freedom	37
Reduced Chi-Sqr	0.25834
Residual Sum of Squres	9.55866
R-Square (COD)	0.99953
Adj. R-Square	0.99948
Fit Status	成功(100)

Figure 37 Fit Curve Data (a) rising part (b) decline part

2) Photodetection Performance Analysis of Ni/n-ZnO/Ni horizontal structure ultraviolet detector under an $Ar\ /\ O_2$ Ratio of 18: 2

From the I-V characteristic curve, it can be observed that the electrode of the Ni/n-ZnO/Ni horizontal structure ultraviolet detector forms an ohmic contact with the semiconductor. In this ohmic contact, field emission, as shown in Fig. 38, predominates, with carrier tunnelling occurring through a narrow potential barrier. The contact resistance between the metal and semiconductor is very low, and there is no significant barrier effect. As a result, free electrons from the metal can directly enter the conduction band of the semiconductor, without causing any signal distortion.

Theoretically, ZnO is a wide bandgap semiconductor with a bandgap of approximately 3.3 eV, which corresponds to a wavelength of around 375 nm. As a result, its effective spectral response is

expected to be limited to wavelengths below 375 nm. However, experimental results show that the ultraviolet photodetector still exhibits responsivity in the 400–440 nm range.

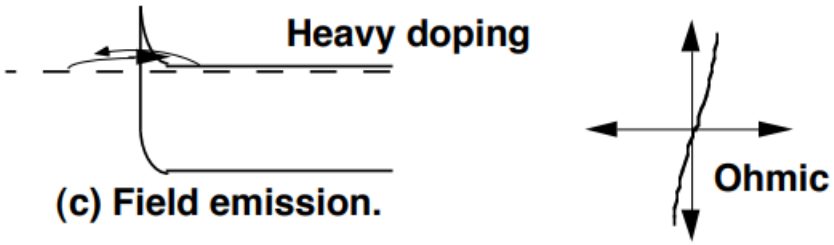


Figure 38 Conduction mechanism of field emission [26]

Defects in solids are unavoidable, especially when physical vapor deposition (PVD) is used to fabricate thin films. In such processes, non-equilibrium growth conditions, limited surface mobility of adatoms, and rapid deposition rates often lead to the formation of structural defects, such as vacancies, interstitials, dislocations, and grain boundaries. In this case, the ZnO thin film is polycrystalline. Furthermore, the film growth time for this experiment was 30 minutes, and the film thickness is estimated to be around 100 nm–150 nm. These factors likely affect the sub-bandgap properties of the material, leading to the detection of photons with wavelengths beyond the typical bandgap energy, thereby extending the spectral response into the 400–440 nm range [30].

To address this issue, a potential solution is to increase the deposition time, adjust the deposition rate, or employ more precise fabrication techniques, such as chemical vapor deposition (CVD). These adjustments can help minimize defects, improve film quality, and reduce sub-bandgap states, thereby enhancing the performance and narrowing the spectral response closer to the theoretical limit.

Another contributing factor could be the lack of proper filtering of the light source. Without appropriate filtering, the light emitted may cover a range of wavelengths rather than being confined to a single wavelength. This broad spectrum could lead to unwanted responses outside the desired detection range, causing deviations from the expected performance.

Higher bias voltages lead to more significant changes in photocurrent. According to Dr. Rüdiger Paschotta, in metal–semiconductor–metal (MSM) photodetectors, applying a bias voltage allows incident light to generate electron–hole pairs within the semiconductor [31]. These charge carriers are then separated and collected by the electric field, resulting in a measurable photocurrent. In semiconductor devices, increasing the bias voltage will shorten the carrier migration time, thereby enhancing the kinetic energy of the carriers. When the electric field strength is high enough, carriers may collide to create new electron-hole pairs, a process called impact ionization. The amplitude of the corresponding current increase.

At the moment of illumination, the Ni/n-ZnO/Ni horizontal structure ultraviolet detector exhibits a fast time response, characterized by a rapid time constant. This swift response is primarily attributed to the drift motion of photogenerated carriers within the depletion region. Under the influence of an external electric field, carriers drift quickly through this region, leading to the initial sharp rise in photocurrent. Following this fast response, a slower decay phase emerges, which is mainly governed by the diffusion of electrons and holes in the neutral region. In this region, carriers move through diffusion rather than drift, and their transport is significantly influenced by factors such as crystal defects, impurity scattering, and material non-uniformities. These effects slow down the carrier movement, resulting in a longer response time associated with the slow time constant [32][33].

Similarly, at the moment the illumination is removed, the Ni/n-ZnO/Ni horizontal structure ultraviolet detector exhibits a sharp drop in photocurrent, indicating a fast response associated with a

short time constant. This rapid change is again attributed to the drift of remaining free carriers in the depletion region, which are quickly swept out under the influence of the external electric field.

Following this immediate response, the photocurrent enters a slower decay phase, primarily dominated by the diffusion and recombination of residual carriers in the neutral region. In this region, carrier movement is no longer assisted by the electric field and instead relies on diffusion, a slower process affected by crystal defects, impurity scattering, and carrier trapping. As a result, the current gradually returns to its dark state, characterized by a longer time constant.

These two phases together constitute a complete cycle of photogenerated carrier transport, enabling the device to seamlessly transition into the next operational cycle.

In the experiment, only various UV-A wavelengths are tested, which decreased its universality for the whole range of ultraviolet wavelengths though the results show the capability for ultraviolet wavelength detect. Including UV-B and UV-C wavelengths in future studies would improve the comprehensiveness of the findings.

3) Problem-Solving Approaches

To summarize the above analysis and the problem, the following approaches are listed.

- **Optimization of deposition parameters**

Improving the crystal quality of the ZnO thin film is crucial for reducing the density of sub-bandgap defect states. This can be achieved by extending the deposition time, lowering the deposition rate, or increasing substrate temperature to promote better adatom mobility.

- **Adopting alternative fabrication techniques**

While physical vapor deposition (PVD) is convenient and cost-effective, switching to more precise fabrication methods such as chemical vapor deposition (CVD), could significantly enhance film uniformity and crystalline quality. These techniques offer better control over film thickness, stoichiometry, and grain boundary formation, thereby reducing defect-induced sub-bandgap absorption.

- **Spectral filtering**

To address the undesired response in the 400–440 nm range, high-precision filtering of the light source may help suppress the contribution from defect states and limit the response to the target UV range.

- **Improving experimental environment**

The experimental environment is a crucial factor influencing the performance of the Ni/n-ZnO/Ni horizontal structure ultraviolet detector. Fluctuations in temperature, humidity, and atmospheric conditions can significantly affect the material properties and device behaviours. To minimize these effects, experiments should be conducted in a controlled environment. Moreover, ensuring a stable and consistent light source during measurements is essential to prevent fluctuations that could introduce noise and inaccuracies. Given the short wavelength and high sensitivity of light waves to vibrations, it is also important to avoid any movement by personnel during the experiment to maintain measurement precision.

VI. Experimental Summary

1) Device Manufacturing and Testing Processes and Key Points

- **Device manufacturing process and key points**

The ultraviolet photodetector investigated in this study was fabricated using sapphire (Al_2O_3) as the substrate and zinc oxide (ZnO) as the target material. ZnO thin films were deposited using the magnetron sputtering technique under an equal mixture of oxygen and argon gases, enabling uniform film growth. Following the deposition, ultraviolet (UV) photolithography was employed. The process utilized a UV lithography machine with a mercury lamp as the primary light source to expose the photoresist and transfer the mask pattern onto the ZnO surface. Proper alignment between the mask and the sample was achieved with the aid of an adjustable microscope, ensuring precise pattern transfer. During alignment, it was found effective to position the sample slightly to the lower right of the center before making fine adjustments.

A critical fabrication step involved the preparation of Ni-based interdigitated electrodes, which was achieved using conventional UV photolithography followed by a lift-off process. This sequence included thorough cleaning, uniform photoresist coating, prebaking, exposure, metal deposition, and stripping. It is essential to control the amount of photoresist applied, as excessive amounts could negatively affect experimental observations. Additionally, the hot plate must be preheated and stabilized before placing the photodetector on it, and the mercury lamp requires a warm-up period before exposure to ensure consistent light intensity.

- **Device testing processes**

The optical properties of fabricated Ni/n-ZnO/Ni horizontally structured photodetectors were evaluated at bias voltages of 0.5 V, 1.0 V, and 1.5 V. The optical properties of the fabricated Ni/n-ZnO/Ni horizontally structured photodetectors were measured by using standard test equipment. The chip was aligned with the detector pins using an electron microscope, and key performance metrics such as photocurrent, dark current, responsivity, response time, and switching behaviours at different scanning voltages were recorded using standard test equipment. These results were then used to qualitatively evaluate the optoelectronic characteristics of the devices. It was observed that the higher the bias voltage, the more pronounced the photocurrent change. Notably, due to the polycrystalline nature of the ZnO film, the recorded data showed wave-like fluctuations, which is consistent with the expected behaviours. External disturbances such as operator movement near the dark box were strictly avoided during testing to minimize interference and maintain data accuracy.

2) Analysis of Device Performance

- **Advantages of the device**

The results from the experimental analysis indicated that the Ni/n-ZnO/Ni UV photodetector exhibited nearly symmetric and linear current-voltage (I-V) characteristics under both illumination and dark conditions. This behaviour confirms the formation of ohmic contact between the metal and semiconductor, ensuring signal integrity without distortion. Moreover, ZnO films fabricated via magnetron sputtering demonstrated favourable optoelectronic properties, including fast photo response and good crystalline quality [34]. The polycrystalline nature of the ZnO film, with a thickness of approximately 100–150 nm, contributed to sub-bandgap states, which extended the detector's spectral response into the 400–440 nm range, beyond the intrinsic bandgap of ZnO. analysis of the device's disadvantages [30].

- **Disadvantages of the device**

Although it is able to realize the basic functions, the UV wavelengths are all in the UV-A range, and there is no detection for shorter waves, so it is hoped that subsequent improvements will increase the universality. Some limitations were identified. The observed cutoff wavelength (~440 nm) exceeded the theoretical value, suggesting a degree of experimental error during testing. The device's responsivity was relatively low, attributed to slow carrier transport during the decay phase, resulting in prolonged response and hesitation times. These factors significantly limit the detector's speed and sensitivity. Additionally, the dark current increased with temperature, introducing additional background noise, lowering the signal-to-noise ratio (SNR), and impeding the detection of weak UV signals [27].

3) Future Research Directions

While this study successfully demonstrated the fabrication and testing of ZnO-based UV photodetectors using traditional magnetron sputtering on sapphire substrates, the current work remains at a preliminary stage due to limitations in experimental facilities and time constraints. Although promising UV detection performance was achieved under laboratory conditions, the device still lags more advanced photodetectors in terms of stability and comprehensive performance evaluation.

- **Suggestions for improvement**

Several avenues can be pursued to optimize the current design and performance. First, extending the deposition time, reducing the deposition rate, or adjusting the substrate temperature can help enhance the crystallinity of the ZnO films and suppress sub-bandgap states. Secondly, employing more precise fabrication techniques, such as chemical vapor deposition (CVD), may yield thinner, more uniform films with superior optoelectronic properties. To reduce spectral deviations, it is advisable to use monochromatic light sources or incorporate optical filters. Finally, conducting experiments in controlled environments—where temperature, humidity, and air quality are stabilized—can mitigate environmental interference. Avoiding operator movement near the setup and maintaining a consistent light source are also crucial for reliable measurements.

- **Research and development outlook**

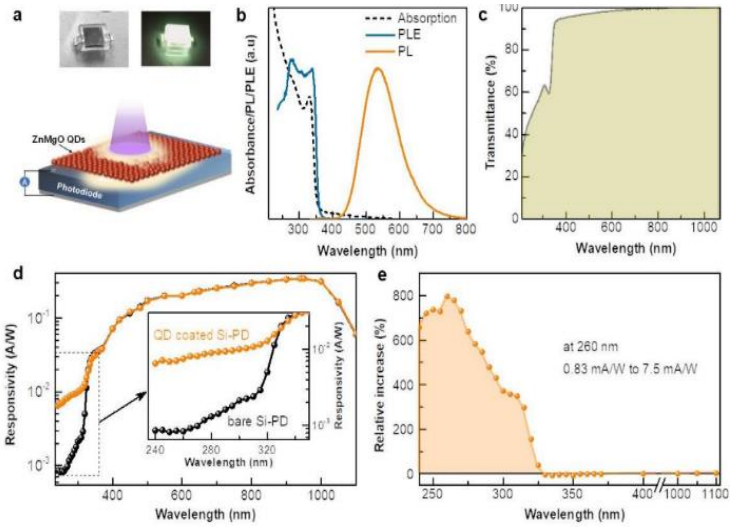


Figure 39 Photophysical properties and performance of QD integrated(120nm) Si-PD [36]

Future advancements in UV photodetectors are expected to focus on enhanced sensitivity, broader spectral coverage, and improved integration with modern optoelectronic platforms. For instance,

integrating biocompatible glutathione (GSH) interlayers in MIS-type devices has been shown to improve carrier separation, device stability, and reduce dark current, offering potential for wearable and bio-integrated applications [35]. Moreover, the use of ZnMgO quantum dots (QDs) as down-conversion layers on CMOS sensors has proven effective as Fig. 39 shows in expanding UV sensitivity and boosting responsivity by up to nine times [36].

Moving forward, scalable fabrication techniques, multifunctional sensing capabilities, and application-driven design—such as environmental monitoring, smart imaging, and photonic edge computing—are likely to shape the next generation of compact, high-performance UV detectors [37].

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